Place-field repetition is what a predictive map must do when self-motion cannot resolve it: a cognitive map represents space only up to the symmetries self-motion cannot break

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Abstract Place cells fire at the same relative location in every identical compartment of a maze, a repetition usually read as the hippocampus failing to tell the rooms apart. We show it is instead what a predictive code must do when its self-motion signal cannot resolve the symmetry, and that the same mechanism sets both the map and the network's capacity to replay. A recurrent network trained only to predict its next observation (***Levenstein et al., 2024***) represents not physical space X but its quotient X/G: it collapses onto a single place every set of positions its own self-motion signal cannot tell apart. The group G is not the environment's symmetry group. It is the subgroup that self-motion fails to break, set by equivariance, not by information. Two head-direction encodings of one bit each, one invariant under a $180°$ rotation and one not, carry the same entropy yet separate completely in the symmetric arena ($p = 1.1 \times 10^{-5}$) and are identical where there is no symmetry ($p = 0.97$); read through the C_4 quotient the code falls to the group-theoretic $1/|G|$ ceiling. Reconstruction, not prediction, is what breaks the account: a matched autoencoder discards the compass, folds every encoding alike ($p = 0.59$), and never generates more than $0.37\times$ waking coverage offline, while one step of prediction recruits the compass, restores the fold, and lets the network dream; weakening the compass then degrades the map and the replay together, in the same order. Map and replay are therefore the same variable, self-motion integration forced by prediction, read out two ways, which is what makes this a unifying account rather than a phenomenon tied to one readout. A learned, unanchored compass builds the fold on its own. The pure predictive code folds completely; a graded symmetry-breaking cue sweep, powered to resolve a genuine information ramp, found no rate-based signature of the partial repetition reported in rodents (***Grieves et al., 2016***; ***Spiers et al., 2015***), only a small chance-exceeding residual already present with no added cue. A cognitive map represents space only up to the symmetries its own motion cannot break.

Introduction

In a maze of visually identical compartments, a place cell fires at the same location in each room: one cell reports several rooms as a single place. ***Spiers et al.*** (***2015***) recorded CA1 place cells in four visually identical compartments arranged in a row off a connecting corridor and found a high degree of spatial repetition. ***Grieves et al.*** (***2016***) then ran the same four compartments in two arrangements: kept in parallel, place cells often repeated fields across compartments; set $60°$ apart on a radial arm instead, repetition largely disappeared, and a separate cohort of animals also took significantly longer to learn to tell the compartments apart in the parallel arrangement than in the radial one, a behavioral cost that tracked the neural one exactly (full description in Results). This repetition is usually read as a limit of the hippocampal map, an inability to tell identical rooms apart. We show the opposite. It is a necessary consequence of learning to predict, and it reveals what a learned spatial code represents when the world is symmetric.

Levenstein et al. (***2024***) showed that a recurrent network trained to predict its own future observations develops spatially tuned units without any spatial supervision, and validated this in an L-shaped arena chosen so that every position has a unique visual fingerprint. But most rodent electrophysiology is done in square arenas, which are rotationally symmetric, and symmetric landmarks make distinct positions observationally identical. This is the classical perceptual-aliasing problem (***Whitehead and Ballard, 1991***), in which distinct world states return the same sensory input, and it raises a question the field has not asked of any learned spatial code: when the world is symmetric, and distinct places look and feel the same, what does the map represent? The problem is not peculiar to one model. The successor representation (***Stachenfeld et al., 2017***), the standard theory linking prediction to spatial coding, inherits it exactly, because positions related by a symmetry of the transition structure have identical expected future occupancy. Existing accounts of the spatial code assume the answer rather than derive it. The successor representation is defined over a state space that is supplied, continuous-attractor models (***Burak and Fiete, 2009***) build the metric in by construction, and boundary-vector models (***Hartley et al., 2000***) read firing from local geometry. None asks what a code *learned* from aliased observations must represent. We ask, and the answer is a quotient: specifically, the *bisimulation* quotient of Markov decision process theory, the coarsest state abstraction that preserves a process's entire future (***Givan et al., 2003***; ***Li et al., 2006***), arising here not from an explicit abstraction step or a reward signal, both absent, but as the state a network converges to when the only thing it is asked to do is predict.

We show that the learned map represents not physical space X but its *quotient* X/G. It writes each orbit of symmetry-related positions as a single place, discarding which copy of the fundamental domain the agent occupies while keeping where it is inside that copy. The group G is not the full symmetry of the environment. It is the subgroup that the agent's self-motion signal *fails to break*. Self-motion is what can, in principle, break the tie: the head-direction system tracks global orientation by path integration and is anchored to landmarks by vision (***Taube et al., 1990***; ***Knierim et al., 1995***), but it breaks a symmetry only when it is not itself invariant under that symmetry. Formally, if the head-direction encoding ϕ satisfies $\phi(h+g) = \phi(h)$ for a symmetry g of the arena, the entire (observation, action) input process is G-equivariant, the posterior over position is G-invariant, and no amount of that signal can lift the ambiguity, so the code must collapse onto X/G. What matters is therefore not the *quantity* of information the compass carries but its *equivariance*, that is, how it transforms under the group. An encoding carrying exactly the same number of bits, but not invariant under g, lifts the fold completely. Head direction is here an *instrument* rather than the subject of the paper: it is the self-motion signal we can manipulate cleanly. The statement is general. Any self-motion signal resolves exactly the symmetries under which it is not itself invariant, and a signal invariant under all of them, such as a pure speed cue, resolves none.

The rest of the paper establishes this quotient law and its consequences. The fold is visible directly in single-cell tuning: place fields grown without spatial supervision repeat at symmetry-related locations when the compass is invariant, once under C_2 and four times under C_4. A controlled experiment that holds information fixed and varies only equivariance separates the two completely, isolating equivariance as the cause. Read through the C_4 quotient the fold produces a parameter-free $1/|G|$ ceiling the networks hit; it is built only under a *predictive* objective, since a matched reconstruction objective folds every encoding alike; and it arises unaided from a learned,

Table 1. The four head-direction encodings. *axis* and *parity* are matched at one bit and differ only in invariance.

encoding	bits	C_2-invariant	C_4-invariant
full (E,S,W,N)	2.0	no	no
parity ({E,S} vs {W,N})	1.0	no	no
axis ({E,W} vs {N,S})	1.0	**yes**	no
const	0.0	yes	**yes**

unanchored compass, surviving one corrupted on a third of its steps. The law then earns its name against data: it explains why place cells in visually identical compartments repeat when the compartments are translationally related and separate when they are rotationally related (***Grieves et al., 2016***; ***Spiers et al., 2015***), and it makes a directional prediction that a published CA1 dataset (***Hockeimer et al., 2023***) is consistent with: recorded in a city-block maze whose alleys run in two orthogonal orientations (40 cells, 5 rats), a cell's repeating fields share their tuning across the translation-related alleys the compass cannot resolve. That dataset does not, on its own, rule out a conjunctive place-by-direction code (Limitations); the decisive test of the same prediction is the model's own, run on the identical geometry before any comparison to animal data (Results). Throughout we distinguish a folded code from a broken one. The folded networks build a reproducible map of the fundamental domain and merely forget which copy of it they are in.

Results

Figure 1 previews the elements this section formalises: the predictive-learning pipeline, the three landmark arenas, the animal phenomenon under explanation, and the quotient map at the centre of the argument.

A design that isolates equivariance from information

Networks receive a $7{\times}7{\times}3$ egocentric view and a five-dimensional SpeedHD action vector, [speed, onehot(h) with $h \in \{\mathrm{E}, \mathrm{S}, \mathrm{W}, \mathrm{N}\}$. We replace the heading block by M onehot(h) for a 4×4 matrix M, preserving dimensionality and column sums (Table 1). Two of the four encodings partition the headings into two classes of two, exactly one bit each, identical entropy, identical activation statistics, and differ only in *which* partition, and hence in whether they are invariant under the $180°$ rotation R^2.

The two hypotheses make opposite predictions. If folding tracks how much information the compass carries, *axis* and *parity* behave alike (Fig. 2a). If it tracks invariance, *axis* folds and *parity* does not.

Predictive learning grows place cells, and symmetry makes them repeat

Trained on prediction alone, the networks develop units with localised spatial firing fields, the place cells reported for this objective by ***Levenstein et al.*** (***2024***) (Fig. 2b). In the asymmetric arena under the full compass, about 480 of the 500 units carry a place field (a median of about two fields per unit). The effect of symmetry is visible in the rate maps before any decoder is applied. When the arena is symmetric and the head-direction encoding is invariant to that symmetry, the fields repeat: a unit in the C_2 arena under *axis* fires at a location and again at its $180°$ image, and a unit in the C_4 arena under *const* fires at all four $90°$ copies. A single unit reports every member of a symmetry orbit as the same place. The sections that follow quantify this and isolate what controls it.

The map collapses onto the symmetry quotient X/G

The fold is a theorem about the inputs before it is a fact about any network.

Theorem (folding onto the quotient)**.** *Let a finite group G act freely on the arena's positions X, with a G-invariant start distribution, a G-equivariant observation likelihood $p(o \mid x)$, and a policy measurable* *with respect to the agent's belief. If the head-direction encoding is G-invariant, $\phi(h+g) = \phi(h)$ for every $g \in G$, then the posterior over position given any history of observations and encoded actions is G-invariant, and any decoder of the orbit element has accuracy at most $1/|G|$.*

Proof. The action of $g \in G$ on trajectories, $\gamma \mapsto g \cdot \gamma$, is a measure-preserving bijection: it fixes the invariant start distribution and the equivariant transition and policy, and it leaves the sequence of observations and encoded actions unchanged, because G-related positions are observationally identical and ϕ is G-invariant. Histories ending at x and at $g \cdot x$ therefore induce the same input sequence, so the network map, a fixed function of that sequence, assigns them the same state, and the position posterior is constant on each G-orbit. No function of the state separates orbit-mates, so an $|G|$-way phase decoder is bounded by $1/|G|$.

The policy hypothesis is concrete, not abstract, and worth cashing out because it is easy to violate by accident. Every trajectory in this paper is generated by a random walk with fixed action probabilities (forward 0.60, turn left 0.15, turn right 0.15, stop 0.10; Methods), the same at every position, heading, and copy of the fundamental domain. Such a policy is a function of nothing the agent's belief could fail to represent, so it is trivially measurable with respect to that belief. The hypothesis is not vacuous: it would fail for a policy that consulted information outside the belief state, for instance a script that explored one room more thoroughly than its g·room image, which would leak room identity into the action stream itself and let a decoder recover it from the policy rather than from the map. Nothing in the experiments does this; the walk cannot tell the rooms apart any better than the compass can.

This is an upper bound, not a prediction. It forbids exceeding the ceiling but does not force the network to reach it, and the free-action assumption (no rotation fixes a cell, satisfied by the even grid) is what gives every orbit exactly $|G|$ members. Two things the paper turns on are therefore empirical, not entailed: that a *predictive* objective drives the code to *saturate* the bound, where a reconstruction objective leaves it far below (next section), and that the folded code is folded rather than broken, since within-domain position survives. The quotient is pinched between a proved ceiling and an empirical floor.

The proof turns on nothing but the sigma-algebra the (observation, action) stream generates, so the ceiling is invariant to everything that sits downstream of that stream. It does not reference the observation's format: a 7×7 pixel patch, a higher-resolution render, or a richer sensory channel are all G-equivariant whenever the underlying arena is, and G-related histories remain indistinguishable whatever the observation's dimensionality. It does not reference network capacity: an arbitrarily large or small decoder computes a function of the same aliased sequence, and no amount of capacity manufactures information the sequence does not carry, so a bigger network is not expected to escape the ceiling, only to reach it faster or more exactly. It does not reference the encoder's identity: any map from the same equivariant input stream inherits the same invariant posterior, whether that map is the field-grown place code studied here, a different architecture, or a hand-built feature extractor. And it survives noise of the kind that leaves the equivariance intact: Eq. 1 states exactly this, since a compass corrupted on 30% of steps (its own subsection) changes only how good an estimate of the true heading each step supplies, not whether the corrupted encoding remains invariant under the group, and the fold does not move. What does lift the ceiling is a channel that breaks the equivariance itself, which is a statement about information content, not about richness, capacity, architecture, or noise level (see the analysis of a candidate symmetry-breaking cue below). The three worries a richer observation, a bigger network, or channel noise might quietly rescue the ambiguity are answered by the same argument: none of them changes the sigma-algebra the group acts on.

Under folding onto X/G the population state satisfies $h(x) = h(R^2 x)$, so no decoder can recover which element of the orbit $\{x, R^2 x\}$ produced it. We therefore train a linear decoder for *orbit phase*, holding out whole orbits across cross-validation folds so the decoder must generalise a phase direction to positions it has never seen. Chance is 0.500.

In the C_2 arena, *axis* and *parity* carry the same entropy (one bit) and the same information

Table 2. Orbit-phase decoding accuracy (chance = 0.500). $n = 10$ per cell for s1 and s2, $n = 8$ for s4.

arena	*full*	*parity*	*axis*	*const*
s1 (C_1, no symmetry)	0.981	0.972	0.971	0.966
s2 (C_2)	0.978	0.955	**0.552**	**0.552**
s4 (C_4)	0.984	0.933	**0.544**	**0.521**

about the current heading, yet they differ by 0.403 in phase accuracy (Fig. 3). What separates them is equivariance: *parity*'s bit is informative about orbit phase (it distinguishes a position from its 180° image), whereas *axis*'s bit is invariant under that rotation and so carries none. The separation is complete: every *parity* network scores above every *axis* network (Mann–Whitney $U = 0$, $p = 1.1 \times 10^{-5}$, the exact two-sided floor at $n = 10$ vs. 10; $p = 3.2 \times 10^{-5}$ after Bonferroni correction across the three arenas). In the C_1 arena, where there is no symmetry to collapse onto, the same two encodings are indistinguishable ($\Delta = 0.000$, $p = 0.97$). The C_1 null is what licenses the causal reading: *axis* does not degrade spatial coding in general, it degrades it exactly when the arena has a symmetry under which *axis* is invariant.

The chance-level reading of the folded codes is not an artefact of using a linear decoder. On the same population states, a k-nearest-neighbour decoder and a multi-layer perceptron also sit at chance for the invariant encodings in the C_2 arena (*axis*: 0.537 and 0.535; *const*: 0.531 and 0.531; $n = 6$), and both recover orbit phase for the non-invariant encodings (*full* and *parity*, every network ≥ 0.90). The orbit-phase information is absent from the folded code, not merely inaccessible to a linear readout.

None of the folded networks has simply stopped representing space. For 104 of the 112 networks, $\max(R^2_{\text{raw}}, R^2_{\text{domain}}) \geq 0.81$: the folded codes still linearly encode position *within the fundamental domain*, they have merely discarded which copy of it the animal occupies (Fig. 4). The eight exceptions are *const* in the C_4 arena, which fails both C_2-based readouts ($R^2_{\text{raw}} = -0.069$, $R^2_{\text{domain}} = -0.589$) not because it has stopped coding space but because it is folded by the *full* C_4 group, which a C_2 decoder cannot see; the next section resolves it. Domain position and orbit phase move independently across every network and encoding (Supplementary Information, Fig. S10): high domain R^2 pairs with either high or chance-level phase accuracy depending on whether the encoding is invariant, which is the coset-vs-phase signature of a fold rather than a dead code.

The quotient is near-exact: the C_4 map hits the $1/|G|$ ceiling

A code folded onto X/G cannot name the element of an orbit, only the coset, so a $|G|$-way phase decoder is bounded above by $1/|G|$. Reading the C_4 arena through its own quotient (four-way phase, chance 0.250) recovers this pattern (Table 3, Fig. 5). *const* is the only C_4-invariant encoding, and the only one to fall to the C_4 floor (0.275, just above 0.250). *axis* is C_2- but not C_4-invariant, and sits near the C_2 ceiling of 0.5 (0.482). The folded values sit within a few percent of their predicted floors rather than exactly on them. A permutation test against the shuffled-label null (single-seed check, $n = 40$ permutations) confirms this is a small but real effect above chance in the C_2 arena (0.531 vs. a null of 0.499 ± 0.005, the exact permutation floor at 40 permutations), not decoder bias; a full multi-seed run is in progress to fix its size precisely. The decisive quantity for the paper's claim is nonetheless which floor each encoding falls to, not the residual's exact size (Discussion).

This also resolves *const* in the C_4 arena. Read through the C_4 quotient it is plainly a live spatial code, its C_4 fundamental-domain R^2 is +0.646 (minimum +0.535 across all sixteen s4 networks), and its four-way phase accuracy of 0.275 sits at the C_4 chance floor. What looked like a dead code under a C_2 readout is a code folded onto a larger quotient than the readout was built to see. We flag this because a G-folded code and a code that has stopped representing space are easy to confuse, and only the second is a failure.

Table 3. C_4 four-way phase decoding in the s4 arena (chance = 0.250).

encoding	invariance	measured	predicted ceiling
full	none	0.918	—
parity	none	0.860	—
axis	C_2 only	**0.482**	0.500
const	C_2 and C_4	**0.275**	0.250

Table 4. The bound of Eq. 1, computed from the arenas with no free parameters, against the trained networks. Every encoding falls to the accuracy its input-distinguishability predicts; the mean absolute error across the twelve cells is 0.031. (We report the bound as a bound, not a fit: the predicted values are near-degenerate, clustered at 1.0 and at the folding floors, so a correlation coefficient across cells would merely register that separation.)

arena	encoding	distinguishable	predicted	measured
s1	*full* / *parity*	1.000	1.000	0.981 / 0.972
s1	*axis* / *const*	0.934	0.967	0.971 / 0.966
s2	*full* / *parity*	1.000	1.000	0.978 / 0.955
s2	*axis* / *const*	0.000	0.500	0.552 / 0.552
s4	*full* / *parity*	1.000	1.000	0.984 / 0.933
s4	*axis* / *const*	0.000	0.500	0.544 / 0.521

The inputs set the quotient; the objective decides whether it is built

None of this requires a trained network to predict. Call two states (x, h) and (x', h') *input-equivalent* if they produce the same 7×7 observation and the same HD encoding. A code is a function of its inputs, so no decoder can separate input-equivalent states, and phase decoding is bounded above by

$$\mathrm{acc}_{\max} = \tfrac{1}{2} + \tfrac{1}{2} \Pr_{(x,h)}\big[\,(x, h) \nsim (R^2 x,\ h + 2)\,\big], \tag{1}$$

the fraction of orbit pairs the input distinguishes. Evaluating the right-hand side on the arenas themselves, enumerating all 324×4 states and hashing their observations, takes no training at all.

Table 4 also settles a number that would otherwise look like a failure. In the C_1 arena *axis* reaches 0.971 rather than 1.000, because 6.6% of states there are *accidentally* observationally identical to their C_2 partner, putting the bound at 0.967. *axis* and *const* sit exactly on it. They were not failing to fold; they were folding on precisely the ambiguous fraction.

The objective selects which bound is reached.

The inequality is an upper bound, not a prediction: a network is free to discard information its loss does not reward. This is also the one manipulation in the paper that could have shown the fold is a property of this architecture rather than of what it is trained to do: hold the wiring, the compass, and the training data fixed and change only the objective from predicting a future observation to reconstructing the current one. If the fold, the ceiling, and the offline replay that tracks it are architectural rather than objective-driven, none of them should move. We tested this by repeating the C_2 sweep varying only the rollout horizon k. At $k = 0$ the target is the network's own current observation, a pure autoencoder with the same architecture and the same compass (verified exactly); at $k \geq 1$ it must predict a future observation. At $k = 0$ the input still separates 1228 of the 1296 states under *full*, so the compass is right there, yet all four encodings collapse to within 0.005 of one another (Table 5), landing on the *observation-only* bound of 0.500; the spread across encodings is 0.0052 at $k = 0$ against 0.427 at $k = 5$ (Fig. 6a). The compass is not absent under reconstruction, it is *discarded*: reconstructing o_t never requires knowing which of two identical-looking places you

Table 5. Orbit-phase decoding in the C_2 arena as a function of prediction horizon k (chance $= 0.500$, $n = 6$ per cell).

k	*full*	*parity*	*axis*	*const*	*axis* vs *parity*
0 (autoencoder)	0.536	0.541	0.539	0.539	$p = 0.59$
1	0.975	0.941	0.545	0.546	$p = 0.00216$
3	0.968	0.947	0.548	0.545	$p = 0.00216$
5	0.978	0.953	0.553	0.551	$p = 0.00216$

occupy, whereas predicting o_{t+1} does, and one step is enough. Everything downstream of this section, the ceiling, the replay, and the compass ablations, is therefore not a property of a network that happens to also predict; it is what a fixed architecture does *because* the objective is prediction, and reconstruction on the identical network is the demonstration of that, not a control incidental to it.

Under reconstruction the full two-bit compass buys nothing: every encoding folds, and all four sit near chance (Table 5). These are folded codes, not broken ones, the minimum of $\max(R^2_{\text{raw}}, R^2_{\text{domain}})$ across all 24 autoencoder networks is 0.858. A single step of prediction restores the effect in full, and extending the horizon to five adds nothing: the dependence on k is a step, not a gradient.

The same step, not a gradient, appears in place-field size. If the prediction horizon behaved like a successor-representation discount, place fields should widen smoothly with k, giving a representation-scale gradient of the kind reported along the hippocampal dorsoventral axis (***Kjelstrup et al., 2008***). Measuring mean field area (*full* encoding, C_2 arena) instead finds a step: 30.6 ± 0.6 cells at $k = 0$, jumping to 38.1 ± 0.5 at $k = 1$ and never moving further (38.5 ± 0.8 at $k = 3$, 37.4 ± 0.1 at $k = 5$). Prediction versus no prediction sets field size once, the same way it sets the fold; how far ahead the network predicts does not additionally scale it, at least across this range of k and at fixed hidden size. This architecture does not, therefore, offer the discount-gradient account of the dorsoventral axis for free; the horizon here behaves as a switch, not a dial.

The reason is that the autoencoder's target is the current observation, which the compass cannot help predict, so the objective gives the network no reason to integrate self-motion and the C_2-symmetric observation stream determines the code. Prediction of a future observation requires integrating self-motion across at least one step, and it is that integration which resolves the symmetry. The control therefore isolates a property of the objective rather than of reconstruction as such: an objective that does not require self-motion integration does not recruit the compass, whatever else it reconstructs.

The same mechanism sets the map and the capacity to replay

A predictive network can be run offline, driven by its own predicted observation, to generate trajectories without moving through the arena (***Levenstein et al., 2024***), and its offline behaviour tracks the fold. Both the coverage of the arena (relative to a waking path of equal length) and the sequenceness of the rollout (the rank correlation between principal-axis position and time, against time-shift and cell-shuffle nulls) step up between $k = 0$ and $k = 1$, over the same range as orbit-phase accuracy, and both fall as the compass is weakened (Fig. 6b). Coverage runs 1.30, 0.80, 0.45 and 0.31 times waking under *full*, *parity*, *axis* and *const*, the order in which the code folds, and sequenceness falls from 0.53 under *full* (near the waking 0.56) to the cell-shuffle floor (0.20) under *axis*. Offline generation therefore depends on prediction and on head direction in the same way the fold does, and emerges at the same horizon.

This is more than a shared horizon; it is the same variable measured twice. The k-sweep that takes the map from the observation-only bound to the C_4 ceiling (Table 5) is the identical manipulation that takes offline coverage from $0.37\times$ waking under the autoencoder to full engagement under one step of prediction, and the *full*/*parity*/*axis*/*const* order in which encodings fold the map is the order in which they lose the replay. A predictive objective does not separately grow a spatial map and a capacity to generate offline activity; it grows one thing, self-motion integration recruited because prediction requires it, and the map and the replay are two readouts of that single mechanism. Reconstruction, the only objective tested here that removes the mechanism, removes both readouts at once and by the same amount. This is the sense in which predictive learning is a unifying account of the spatial code rather than a phenomenon confined to place-field statistics: not that it produces a map and, separately, produces replay, but that the one thing it does, recruit self-motion because the objective requires it, is read out as both.

Four things would have shown this is a phenomenon specific to one setup rather than a theory of what a predictive code represents, and each was tested rather than assumed. If the fold tracked how much information the compass carries rather than its equivariance, *axis* and *parity* would behave alike; they do not, separating completely when matched at one bit each ($p = 1.1 \times 10^{-5}$) while being indistinguishable in an arena with no symmetry to fold onto ($p = 0.97$). If the fold were a property of the input pipeline rather than the objective, changing only the objective while holding the wiring, the compass, and the training data fixed would not touch it; it does, collapsing every encoding to chance under reconstruction and restoring the ceiling after one step of prediction. If the ceiling were a property of network capacity or architecture, the proof establishing it would need to reference the decoder or the hidden size; it does not, turning on nothing but the sigma-algebra the observation-and-action stream generates, so no amount of capacity is expected to lift it, a theoretical guarantee that a hidden-size and arena-size sweep would test empirically but that sweep has not yet been run (Limitations). If offline replay were independent of the compass, weakening or removing it would leave replay unaffected; instead coverage and sequenceness fall in the same order the map folds, and hit the same floor the autoencoder hits when the objective, not the compass, is removed. Each of these is a way the account could have come apart, checked against the network's behaviour rather than assumed by it.

Sequenceness is scored as |Spearman$(x(t), t)$|, an absolute value, so it measures whether a rollout is an ordered sweep at all and is blind by construction to whether that sweep runs forward or in reverse; a principal-axis projection also has no privileged sign without an external reference direction (the preceding wake heading) to compare it against, which this statistic does not use. We therefore make no claim here about forward versus reverse replay. Real reverse replay is reward-triggered (**Foster and Wilson, 2006**; **Diba and Buzsáki, 2007**), and this network has no reward signal, so the account predicts its offline sweeps should not show the reverse, reward-linked form; testing that prediction needs a directional statistic referenced to the immediately preceding trajectory, which we have not built.

The dissociation survives a noisy compass

The head-direction signal used so far is a noiseless oracle, whereas the biological compass drifts and is re-anchored by vision. To test whether the fold depends on a perfect signal, we retrained the C_2 networks with the reported heading corrupted: on 30% of steps the compass was rotated by a random $\pm 90°$ before encoding. The dissociation is preserved. Orbit-phase decoding is 0.556 under *axis* and 0.922 under *parity*, a separation of 0.367 that remains complete: every *parity* network scores above every *axis* network (Mann–Whitney $U = 0$, $p = 0.0022$ at $n = 6$ vs. 6). Relative to the clean networks the separation narrows only slightly (from 0.403), driven by a small drop in *parity* resolution ($0.955 \rightarrow 0.922$) while the folded *axis* code is unchanged ($0.552 \rightarrow 0.556$). A compass corrupted on a third of steps still folds the same symmetries it does when clean (Fig. 7a). (These networks were trained for $30{,}000$ steps rather than $80{,}000$; the effect is already complete by then.)

A learned, unanchored compass builds the quotient unaided

The strongest form of the test removes the oracle entirely. Rather than degrade a supplied compass, we withheld absolute heading and gave the network only *angular velocity*, the per-step turn, so it had to integrate its own heading, unanchored, as a path-integrating head-direction system does. An integrated compass is defined only up to a global rotation, so it is invariant under any rotation of the arena and cannot, by itself, separate a position from its symmetry image.

The learned compass reproduces the fold from first principles. In the asymmetric arena the network decodes orbit phase almost perfectly (0.971, $n = 10$), because the arena's unique landmarks anchor absolute position. In the C_2 arena it falls to chance (0.526, chance 0.500, $n = 10$), folding the two halves onto one map. Raw position is not recoverable ($R^2 = -0.08$) while position *within* the fundamental domain is ($R^2 = 0.89$): the network knows where it is inside a copy, not which copy. In the C_4 arena it folds likewise (0.523, $n = 8$; $R^2_{\text{domain}} = -0.47$, the four-fold signature a C_2 decoder cannot see). This is the same fold the hand-designed *axis* and *const* encodings produce, now arising from the compass itself (Fig. 7b). The fold is therefore not an artefact of imposing an invariant encoding: it is the generic consequence of a self-motion signal that carries no absolute reference under the arena's symmetry, the situation of a biological compass that has lost its landmark anchor. This is the quotient law in its purest form. A path-integrated compass is defined only up to a global rotation, so the angular-velocity channel is invariant under every arena rotation *by construction*; the code then folds precisely when the *observation* channel is invariant as well, which is why the same learned compass folds the symmetric arenas but lifts the fold in C_1, where unique landmarks anchor absolute heading. The fold is therefore not imposed on the network: it is what an unanchored self-motion system must do wherever its observations are also aliased. The hand-designed encodings are a controlled way to dial that invariance; the learned compass shows the invariance is where a real self-motion signal already lives.

Two natural readouts fail, in two different ways

This is a completeness check on our own earlier analysis of these networks, not a critique of prior published work: an earlier pass through this project used rotational autocorrelation as its readout, and we show here why that metric had to be replaced rather than merely supplemented. Decomposing each unit's rate map into the four characters of C_4 gives an isotypic power spectrum $P_0 \ldots P_3$. Rotational autocorrelation is exactly RA $= P_0 - P_2$, and it is therefore *blind to C_2 structure*: invariance under R^2 annihilates the odd characters P_1 and P_3 while leaving P_0 and P_2 untouched, so a perfectly C_2-folded code and pure noise both read RA ≈ 0. Empirically, on a clean 17-network sweep, RA cannot separate s2 from s1 ($p = 0.056$), whereas odd power $P_1 + P_3$ separates them completely ($p = 0.0079$, the exact floor at $n = 5$ vs. 5; Fig. 9b,c).

Odd power, however, is itself confounded. Under *axis* it falls in the C_1 arena too, where there is no symmetry: the drop odd(*parity*) − odd(*axis*) is 0.095 in s1, 0.111 in s2 and 0.112 in s4, so 85% of the s4-sized effect is present where nothing can fold, and it separates completely in s1 ($p = 1.8 \times 10^{-4}$), a confident report of a fold that does not exist. Odd power conflates genuine collapse onto the quotient with the encoding merely reshaping the rate-map spectrum. Orbit-phase decoding is immune to that and returns a clean null.

We therefore report phase decoding as the primary readout and the isotypic spectrum as a secondary, descriptive one, kept in the paper because the same superseded readout would have supported the same qualitative conclusion by an argument that does not hold, and a reader reconstructing this analysis from scratch should not rediscover that dead end.

The quotient map is reproducible, not degenerate

Because the folded encodings decode orbit phase near chance (about 0.55, reliably just above the 0.5 floor but far below the ~ 0.95 of the non-invariant encodings), we asked whether they had learned a spatial map at all, or whether ablating the compass had degraded the representation more broadly. Two measures argue that the map is intact (Fig. 8c,d). Spatial representational similarity, the Spearman correlation between neural distance and Euclidean distance across pairs of positions, fell as the compass was ablated, from 0.70 under *full* to 0.32 under *const* in the C_1 arena, and lower still in the symmetric arenas. This is the expected consequence of folding: a position and its group image sit on top of one another in the neural space while lying far apart in the arena, weakening the correlation between the two distances. The correlation between the position-conditioned codes of independently trained networks behaved oppositely and stayed above 0.96 under every encoding in every arena, and 448 of the 500 units retain a place field under *const* in the C_1 arena, falling to about 318 in the C_4 arena. A folded code is therefore reproducible rather than degenerate, and the compass governs how the map is laid out rather than whether one forms. The claim also has a direct topological reading: an Isomap embedding of the population code shows that under *axis* a position and its $180°$ image sit, on average, at 0.17–0.18 of a random pair's distance apart, against 0.85–0.89 under *full* (complete separation across five seeds each, $p = 0.004$), so the folded manifold is the glued fundamental domain the Theorem predicts, not the unfolded arena (Supplementary Information).

The collapse can also be read in the metric used to score remapping in the hippocampus. The population-vector correlation between a position and its symmetry image (***Leutgeb et al., 2005***) — near one when the two share a representation, near zero when the code separates them, is 0.98 under the invariant encodings in the C_2 and C_4 arenas and about 0.28 under *full*, with the matched non-invariant encoding intermediate (*parity*, 0.58 in C_2; $n = 10$ per encoding; Fig. 9a). By this measure the network fails to remap between symmetry-related locations exactly when the compass is invariant under the symmetry, the two copies are written as one place, whereas with a full compass they are separated as strongly as unrelated positions. The same invariant encoding gives a weaker correlation in the C_1 arena (0.71 for *axis*), where there is no arena symmetry to complete the fold.

Folding multiplies place fields and imprints the arena's symmetry on each map

The fold is visible cell by cell, not only in the population decoder. We characterised every unit's firing-rate map and counted its place fields (Fig. 8a). Fields per unit rise with both arena symmetry and encoding invariance, from 1.78 under *full* in the C_1 arena to 3.39 under *const* in the C_4 arena. The per-unit field-count distribution in the unfolded baseline (*full*, C_1) is under-, not Poisson-, dispersed (Fano factor 0.52; χ^2 against a Poisson of the same mean is decisively rejected), concentrated at one or two fields more tightly than a Poisson process predicts; large real environments are reported to show closer to Poisson statistics (***Rich et al., 2014***), so this is a respect in which the model's field-count statistics do not match this literature. On its own the raw count is also a confounded readout: it rises even in the C_1 arena, where nothing can fold (2.44 fields under *const* against 1.78 under *full*), because a more invariant encoding yields more multi-peaked units regardless of symmetry. The fold-specific question is whether the extra fields sit at group-related positions. We measure this as the correlation between a unit's rate map and its own $180°$ rotation, a per-cell index of C_2 symmetry (Fig. 8b). The index saturates near 0.97 exactly when the encoding is C_2-invariant and the arena is symmetric (*axis* and *const* in the C_2 and C_4 arenas), while the matched non-invariant encoding stays low (*parity*, 0.36 in C_2) and *full* stays low throughout (≤ 0.16). An invariant encoding alone induces partial symmetry even in the C_1 arena (0.61 for *axis*), which the arena's own symmetry then completes. The four-fold version of the index reaches 0.97 only for *const* in the C_4 arena, the single fully C_4-invariant condition. Each map acquires precisely the symmetry the compass cannot break.

Scored by the criteria applied to real recordings, the trained population is a realistic mixture rather than a bank of identical units: 76–96% of units qualify as place cells (Skaggs spatial information above threshold) and about 30% are border-modulated (a third or more of their rate-map mass in the outer ring), a composition that is stable across encodings and arenas (Supplementary Information). Folding therefore multiplies fields and rearranges where they sit without changing the cell-type makeup of the map. The multi-field, spatially repeating cells it produces are the in-silico counterpart of the repeated place fields recorded in symmetric and multicompartment environments (***Grieves et al., 2016***; ***Derdikman et al., 2009***).

Place-field repetition is the signature of the quotient

Head direction should resolve rotational but not translational symmetry, and this can be tested in the geometry used to study repeated place fields in rodents. We built two closed 6×6 compartments joined by a corridor and gave them identical interior landmarks, so that the two rooms produced the same set of egocentric views; across all 144 combinations of location and heading the observations coincided exactly. In one arrangement the rooms were related by a translation, under which the compass reads the same in both; in the other by a rotation, under which it does not. We trained networks on each arrangement and, for every network, decoded which room the animal occupied and measured whether individual units repeated their firing fields between the rooms (Fig. 6c).

The two arrangements gave opposite results ($n = 8$ networks each). Under translation the fields repeated almost exactly across the rooms (repetition 0.997) and the room could not be decoded even by a nearest-neighbour classifier tested on cells it had already seen (0.515, chance 0.5), so the network folded the two identical rooms onto one map. Under rotation the repetition vanished (repetition -0.080) and the room decoded perfectly (0.998). Position within a room decoded well in both arrangements ($R^2 = 0.95$ and 0.79), so neither network had stopped representing space. Every translation network separated from every rotation network on all four measures (Mann–Whitney $p = 1.6 \times 10^{-4}$, the exact two-sided floor at $n = 8$ vs. 8). ***Grieves et al.*** (***2016***) reported this same contrast in CA1, and here it follows from how the compass transforms under the room's relation to its partner, translation-invariant or rotation-equivariant (Methods; Limitations). The same dissociation is visible directly in the population geometry, with no decoder involved: an Isomap embedding of the wake population state lands room B almost exactly on top of room A under translation and pulls them an order of magnitude further apart under rotation (Supplementary Information, Fig. S12).

The fold scales to the full four-compartment maze that ***Spiers et al.*** (***2015***) recorded. We built four identical closed rooms in a row, each the translation of the last, off a shared corridor, with every view identical across the four rooms at matched local position and heading (verified exactly: zero pixel mismatch over every room-interior cell and heading, Methods). A predictive network ($n = 8$) folded all four onto a single map: the room could not be decoded above the 0.25 chance of four alternatives, neither by a generalising decoder (0.262) nor by a nearest-neighbour classifier tested on cells it had already visited (0.268), while fields repeated across the four rooms almost perfectly (repetition 0.994) and within-room position stayed fully intact ($R^2 = 0.96$). Room identity is decoded only from these room-interior states (Methods); the connecting corridor is excluded from the readout for the reason given in Limitations. The two-room dissociation is thus not a small-n artefact: at the rodent room count, translation-related identical rooms collapse completely, the "high degree of spatial repetition" ***Spiers et al.*** (***2015***) observed.

The account makes a further prediction, which we test in the model before asking whether it is visible in real neurons. A city-block maze whose alleys run in two orthogonal orientations relates same-orientation alleys by translation (the compass reads the same) and orthogonal alleys by a $90°$ rotation, so a repeating cell's fields should share a directional preference across same-orientation alleys and not across orthogonal ones. A predictive network trained on this maze confirms it: the directional index of same-orientation field pairs is positively correlated ($r = 0.62$; Fig. 10b), and this is the model's own prediction, not a fit to animal data. We then asked whether the signature is visible in real neurons, reanalysing the CA1 recordings of ***Hockeimer et al.*** (***2023***). It is present, at a smaller magnitude ($r = 0.23$; 127 pairs, 40 cells, 5 rats), but the dataset cannot separate it from a conjunctive place-by-direction code that would produce the same correlation without any folding; two controls make the ambiguity explicit (Supplementary Information, Fig. S9). We therefore report the CA1 correlation as *consistent with* the quotient but not diagnostic of it, and treat the model prediction, not the animal reanalysis, as the load-bearing result of this section.

Discussion

The learned cognitive map is a quotient. When the environment has a symmetry the agent's self-motion cannot break, the predictive code writes each orbit of symmetry-related positions as a single represented place, and what it maps is X/G rather than X. This is a statement about group actions, not about information channels: the quotient group G is fixed by the self-motion signal, which resolves exactly those symmetries under which it is not itself invariant. Head direction is blind to translation, translating the world does not change a compass reading — and therefore cannot disambiguate translationally repeated compartments, whereas it is not blind to rotation and can disambiguate rotationally related ones. What the animal, and the network, ends up representing is not the room but the room modulo the symmetries its own motion cannot feel.

The principle is not about head direction. Any self-motion signal resolves exactly the symmetries under which it is not itself invariant: a compass is blind to translation and so cannot separate translationally repeated compartments; a pure speed signal is blind to everything but distance travelled and resolves almost nothing; a place-modulated efference copy would be blind to nothing and fold no arena. Which symmetries a learned map collapses is therefore a property of the agent's sensorimotor access to space, not of the hippocampus. Head direction is simply the access we could vary cleanly here.

The dissociation exists in the animal data

Spiers et al. (***2015***) recorded CA1 place cells in four visually identical compartments arranged in a row off a connecting corridor, and found "a high degree of spatial repetition with a slight degree of rate-based discrimination". Those compartments are translationally related, and the authors conclude that "path integration is not used, or at least not spontaneously" to separate them. ***Grieves et al.*** (***2016***) then ran the same four compartments in two arrangements. Parallel, all facing the same way and therefore translationally related: "place cells often exhibited repeated fields". Radial, at $60°$ to one another and therefore rotationally related: "significantly less place field repetition was apparent" (all $p < 0.0001$). The dissociation survived removal of the orientation landmark, so it is not carried by extra-maze vision, and the authors attribute it to "directional information derived from the animal's self-motion", hypothesising the head-direction system as the source.

The same study also reports a behavioral consequence of the fold, not only a neural one. In a separate cohort, rats learned a novel odor–location discrimination that required telling the four compartments apart. Learning "was significantly impaired" in the parallel arrangement relative to the radial one: fewer animals reached the full discrimination, and those that did needed many more sessions to criterion (***Grieves et al., 2016***). Grieves and colleagues conclude that "place field repetition constrains spatial learning." This is a stronger claim than a correlate in firing rates: it is the prediction that a folded code should have a behavioral cost, tested directly, in the same paper, with the same manipulation.

Our result supplies the reason for both halves of that study. A compass is *invariant* under translation and *equivariant* under rotation. The input process is therefore G-equivariant in the parallel arrangement and not in the radial one, so a predictive code must fold onto the quotient in the first case and may lift it in the second. Repetition is not a failure of the hippocampus; it is what a predictive code must do when its self-motion signal cannot see the symmetry.

Two things follow that the animal data cannot establish. First, the effect is a property of *invariance*, not of how much the compass tells you: two encodings carrying the same single bit dissociate completely (Table 2). Second, it is a property of *prediction*: under a matched reconstruction objective the compass is inert (Table 5), so head direction cannot disambiguate compartments unless the objective forces the network to integrate it. Whether a purely reconstructive circuit would show repetition even in the radial arrangement is, to our knowledge, untested in animals.

The account also yields a lesion prediction, with a dose attached. Lesions of the anterodorsal thalamus degrade the head-direction signal and destabilise hippocampal place fields (***Calton et al., 2003***). We predict that degrading the compass should bring field *repetition* into the *radial* arrangement, where intact animals show little, because without direction rotation joins translation as a symmetry the animal cannot resolve. The robustness control sets the threshold high, though: in the model a compass corrupted on 30% of steps still resolves the fold, and full repetition appears only in the *const* limit of near-total directional loss (which folds the C_2 arena to near chance, Table 2). A partial ADN lesion should therefore produce partial repetition, graded by how much of the head-direction signal survives.

The account also predicts an experience-dependent *unfolding*. ***Carpenter et al.*** (***2015***) found that the grid code across two connected compartments begins repeated and becomes a single global representation with experience. In our terms the repeated code is the quotient, and it unfolds once the animal acquires any cue (a faint distal landmark, a learned context) that breaks the symmetry the compass could not. The fold is therefore a starting point set by self-motion, not a fixed endpoint: a weak symmetry-breaking channel should, given time, lift it, while a fully symmetric one cannot.

What the model reproduces, and what it does not

The two rows marked *absent* are architectural, not empirical: head direction enters this network as an oracle input rather than being learned, and no non-negativity constraint is imposed, which is the condition under which hexagonal grids emerge in trained path integrators (***Sorscher et al., 2023***). Nothing here bears on the ring or toroidal manifolds of entorhinal cortex, and we make no claim about them.

The row marked *tested, not found* is the one place the model and the animals part on the fold itself, and we tested it directly. Spiers reported repetition *with* a slight rate-based discrimination, and Grieves reported purely local remapping: the animals keep, in the rate channel, some trace of which compartment they are in. Our network, given perfectly identical rooms, keeps none, and folds completely. We reasoned that real compartments are never perfectly identical, so a cognitive map should not collapse exactly onto X/G but toward it, up to a residual set by whatever faint cue does distinguish the rooms, and set out to test whether a symmetry-breaking channel too weak to lift position would still modulate rate, producing repeated fields that carry a consistent rate difference between compartments (Spiers' signature), with the complete fold reported here as the zero-cue limit of a graded phenomenon.

Eq. 1 lets that curve be predicted before any network is trained, by evaluating the same input-distinguishability bound on the cued arena. A first attempt, a uniform brightness shift on room B's egocentric view of amplitude ε, added to every pixel and clipped to the valid range, turned out to give a step rather than a ramp: below the one-part-in-255 rounding threshold ($\varepsilon < 1/510$) the cue changes no pixel and the input is exactly as aliased as at $\varepsilon = 0$ (bound 0.500); at or above it, every cued state becomes distinguishable from its uncued image and the bound jumps to 1.000. The amplitudes 0.05–0.4 used in that attempt all sat at the ceiling, not on a ramp between the two floors, so we replaced it with a cue built to vary smoothly in the information it carries: fixed per-pixel Gaussian noise ($\sigma = 0.05$) competing against a room-B tint swept in float precision to target single-step discriminabilities $d' = \varepsilon\sqrt{N}/\sigma$ evenly spaced over $\{0, \dots, 3\}$ (`environments/arena.py`, `experiments/run_multi.py`). This puts the bound itself on a ramp, and training confirms it: both a generalization and a within-trajectory room decoder rise smoothly and monotonically with d' (Spearman $\rho = 0.92$ and 0.99, $n = 64$, $p < 10^{-25}$; $d' = 0$ vs. $d' = 3$, exact two-sided Mann–Whitney U, $p = 0.000155$, $n = 8$ vs. 8), so the sweep now tests the prediction as designed.

The prediction itself is not borne out. Field-shape repetition stays at ceiling across the entire sweep (≈ 0.997, $p = 0.88$ comparing $d' = 0$ to $d' = 3$, $\rho = 0.17$, n.s.), and a rate-asymmetry readout built for this test (the median relative firing-rate difference between rooms, computed on the top-100-variance units with no position matching) is flat throughout ($p = 0.88$, $\rho = 0.02$, n.s.): we find no evidence this measure picks up a rate-based signal at all, let alone one that leads position as the prediction required. Room decodability itself rises with d' but stays well below the conservative single-observation Bayes ceiling at every point tested (e.g. at $d' = 3$, generalization accuracy 0.59

against a ceiling of 0.93 for $T = 1$), consistent with information being present in the input but not fully recruited by the network, rather than recruited into a graded rate code. This is an honest negative on the specific rate-leads-position prediction, as pre-registered. Reconciliation with Spiers' partial fold has to come, in this architecture, from the residual already present at zero added cue (generalization room decode 0.510, itself significantly above chance, Wilcoxon $p = 0.023$, $n = 8$) rather than from a graded rate-versus-position dissociation: some trace of compartment identity survives even the "perfectly identical rooms" limit, but it does not grow into the kind of rate code Spiers observed as the cue is strengthened.

The objective control constrains the mechanism further: a matched autoencoder with the same wiring and the same compass folds every encoding alike ($p = 0.59$), which rules out the fold being a property of the input pipeline rather than of the learning. It is an objective that requires predicting a future observation, not reconstructing the present one, that recruits the compass.

Relation to other models of the spatial code

The parallel-versus-radial dissociation is a discriminating test among models of place-cell firing, because the two arrangements are built from the same compartments and are locally identical in both. A model that fixes firing from local sensory or boundary geometry alone must therefore predict the same repetition in each: the boundary-vector account (***Hartley et al., 2000***), in which a field is a fixed function of the distances to walls, gives identical fields wherever the local geometry recurs, parallel or radial. A reconstruction objective behaves the same way for the same reason, and our autoencoder control confirms it: with the compass held identical it folds every encoding alike. Continuous-attractor path integrators (***Burak and Fiete, 2009***; ***Sorscher et al., 2023***) build the metric in by construction and do not speak to what happens when observations alias. The successor representation (***Stachenfeld et al., 2017***; ***Momennejad et al., 2017***) is predictive, but over a state space that is supplied rather than inferred from aliased observations, so whether two symmetric compartments are one state or two is an assumption, not a result (Table 7). Read the other way, though, our result *instantiates* the successor representation rather than competing with it. Under G-equivariance, with a G-invariant prior and a policy measurable with respect to the aliased belief, the agent's belief process (its posterior over position given the observation and action stream) descends to a Markov process on X/G, because G-related states are belief-indistinguishable. This is the symmetry-induced Markov decision process homomorphism studied in reinforcement learning (***Givan et al., 2003***; ***Ravindran and Barto, 2004***; ***van der Pol et al., 2020***), and the equivalence it induces, G-related states are collapsed exactly because they agree on the entire future observation-and-reward distribution, is a bisimulation relation (***Ferns et al., 2004***): the coarsest state abstraction that preserves an MDP's dynamics (***Li et al., 2006***), computed here not by an explicit bisimulation metric but as the by-product of a network trained only to predict. A predictive learner that acquires the successor representation of its *belief* state should therefore represent the bisimulation quotient X/G, giving a concrete, verified instance of a claim usually established by construction (***Castro, 2020***) arising instead from a self-supervised objective with no reward signal, no explicit abstraction step, and no group structure given to the network. The fold is not a departure from the SR account but its consequence once the state space is inferred from aliased experience rather than supplied.

The residual is where this equivalence becomes exact rather than approximate. Table 4's input-level bound is exactly 0.500 for *axis* and *const* in the C_2 arena (distinguishable fraction 0.000): unlike the C_1 arena, where 6.6% of states are accidentally observationally distinct and set the floor above chance, there is no leakage available at the input for the network to exploit here. The few-percent residual above that floor (this section) cannot, therefore, be a bisimulation relation that is merely near-exact at the observation level; the observations are exactly bisimilar. What is approximate is the network's realisation of that equivalence: a trained SR is a fixed point of temporal-difference learning that gradient descent approaches but does not reach exactly, so a finite-capacity network trained for a finite number of steps is expected to sit near, not on, the invariant manifold the exact quotient defines, and the size of that gap is a property of training and estimation, not of the bisimulation relation itself. The near-exact floor is therefore the signature of a network computing an approximate successor representation of X/G, sitting close to the quotient ceiling rather than the un-quotiented one (~ 0.95, Table 2); a partially broken fold and an approximately-converged exact one are observationally similar only in being imperfect, and the input-level bound is what tells them apart.

Two learned-map models place sharply against the same test. The clone-structured cognitive graph (***George et al., 2021***) disambiguates aliased observations by *transition context*, cloning a latent state whenever the routes through it differ; in the parallel compartments the two rooms are entered by distinct paths, so a clone graph should *split* them and predict no repetition — the opposite of what ***Spiers et al.*** (***2015***) recorded. It is thus a falsified competitor here, not an open alternative. The Tolman–Eichenbaum machine (***Whittington et al., 2020***), by contrast, factorises a structural code that is a function of the action sequence; an action (self-motion) code invariant under the arena's symmetry folds that structural code just as ours folds, so the two accounts agree on the direction of the effect and differ only in mechanism.

What predictive learning adds is a derivation. Trained only to predict its next view from self-motion, the network folds or lifts the identical compartments according to whether the compass is invariant under the arrangement's symmetry, reproducing the animal dissociation from a single group-theoretic principle rather than fitting it. The contribution is therefore not that prediction grows place cells (***Levenstein et al., 2024***), but that it specifies *when* the spatial code collapses under symmetry and when it does not, a question the other model classes leave open.

Limitations.

A first group of limitations are architectural absences rather than failures of the account, on the same footing as the missing grid torus. The arenas are discrete 18×18 grids with four headings, so the head-direction "ring" has four points and no continuous ring topology; nothing here speaks to the ring or toroidal manifolds reported in entorhinal cortex (***Gardner et al., 2022***). The networks develop place-like but not grid-like tuning (gridness over the 100 most spatially tuned units: mean -0.35, maximum $+0.08$, none above the conventional 0.3 threshold), expected given that head direction is supplied as an input rather than learned and that no non-negativity constraint is imposed (***Sorscher et al., 2023***); the toroidal code of ***Gardner et al.*** (***2022***) cannot exist in this model, and we do not claim it. The architecture has no membrane dynamics or spikes and updates once per behavioural timestep, so it has no theta oscillation, phase precession, or theta sequences, and no excitatory–inhibitory separation. All networks are trained once on a fixed arena and evaluated shortly after, so the model speaks to the representation a predictive objective converges to, not to whether it drifts across sessions once training has converged (representational drift), and all results use a single architecture (hidden size 500, horizon $k = 5$); the horizon sweep shows the fold from $k = 1$ and the initialisation study varies the recurrent weights, but hidden size and arena size are not swept.

A second group are empirical properties of the trained networks that depart from biology to varying degrees, most of them already bounded by a control reported in Results. The head-direction signal is supplied rather than continuous, but this is not left untested: a control corrupting the compass on 30% of steps leaves the dissociation intact, and a *learned* compass integrated from angular velocity alone reproduces the fold on its own (Results); what is not modelled is a continuous compass that drifts and is re-anchored by vision over time. Real open-field place cells are largely omnidirectional (***McNaughton et al., 1983***; ***Markus et al., 1995***); this model's units are only partly so (opposite-heading rate-map correlation, *full* encoding, C_1 arena: mean 0.56, median 0.64, 35% of unit pairs above 0.8, the range a strongly omnidirectional cell would occupy), because head direction enters as a direct input rather than being merged away by path integration upstream of place cells as it is thought to be biologically. A proposed developmental ordering, in which population-manifold topology crystallizes before metric geometry, does not hold here under a shuffle-null estimator (position is linearly decodable at $R^2 = 0.79$ before training, and loop topology is recovered only through a nonlinear embedding, `topology_robustness.csv`), so we make no topology-before-geometry claim. Place fields do not anticipate future position (Table 6); the discrete one-cell-per-step grid would not resolve a sub-cell anticipatory shift, so this bounds rather than rules out a very small lead.

A third group are open items that bound how far the results generalise. Symmetry here is a property of the landmark layout, not the arena geometry, which is square and therefore C_4-symmetric in every condition, so only the $C_1/C_2/C_4$ rotational arenas satisfy the Theorem's hypotheses exactly. The compartment mazes are not, strictly, an instance of the Theorem's finite group action, because the corridor joining translation-related rooms lacks their cyclic closure; a temporal-conditioning analysis rules out the natural alternative that the two-room dissociation is corridor memory rather than head-direction equivariance (Methods). A more decisive control, mirroring Table 5's sweep for the rotational arenas, asks whether room decodability on the compartment maze rises with k, which would show the residual fold is horizon-limited rather than architecture-limited. We have run this for $k \in \{0, 1, 3\}$ ($n = 8$ networks each): steady-state room decodability (more than ten steps after room entry, to exclude the corridor-memory transient) stays at chance at every horizon tested (generalising decoder 0.508, 0.502, 0.510; none individually distinguishable from chance, Wilcoxon $p = 0.25$, 1.00, 0.055) with no trend across them (Spearman $\rho = 0.07$, $p = 0.73$, $n = 24$), while repetition remains near ceiling throughout (0.98–0.99; Methods). This supports architecture-robustness over horizon-limitation for the range tested, but $k = 5$ and $k = 10$, the longer horizons where an objective would have the most reason to recruit a weak corridor cue, remain untested, so a late-emerging horizon dependence cannot yet be ruled out. The one real-neuron test in this paper, the CA1 city-block reanalysis (Results; Supplementary Information), is similarly exploratory rather than diagnostic: a conjunctive place-by-direction code predicts the same directional-index correlation without any folding, two controls confirm the two accounts are not separable in these data, and we treat the model's own prediction, not the animal reanalysis, as the section's evidence for that reason.

Outlook

The broader appeal of predictive learning is that a single, domain-general objective, predict the next observation, grows structured spatial representations with no hand-built spatial prior, in artificial networks as in the hippocampus (***Levenstein et al., 2024***; ***Stachenfeld et al., 2017***). What this study adds is that the geometry of the learned code is then set by the symmetry structure of the agent's interaction with the world, the group action of the environment on the self-motion signal, rather than by the architecture. That makes the paradigm a controlled developmental probe. Because the representation is learned rather than wired in, each factor that shapes it in an animal, and is there confounded or fixed, becomes a variable that can be turned on its own: the geometry of the enclosure, including irregular and non-convex ones; the information content and invariance of the self-motion signal, as here; the locomotor policy and the trajectory statistics it induces; and the task the agent is trained under. Charting how the code changes along these axes is a way to ask which features of an environment and an agent are responsible for which features of the representation.

Architecture is a variable of the same kind. The present network receives head direction as an oracle input and places no non-negativity constraint on its units, and it accordingly grows place-like but not grid-like tuning; hexagonal grids emerge in trained path integrators precisely when those constraints are imposed (***Sorscher et al., 2023***). Progressively more biological designs, a head-direction signal that is learned and allowed to drift rather than supplied, non-negative rates, separate path-integration and sensory streams, and modules for planning — would let one ask which ingredients are necessary for the joint emergence of place and grid cells and for the developmental order in which they appear. The aim of such a program is to turn the catalogue of spatial cell types into a set of consequences that follow from an objective, an architecture, and an experience, of which the symmetry-driven fold reported here is one.

Methods

Architecture and training.

We use `pRNN_th` (***Levenstein et al., 2024***) unchanged: a layer-normalised recurrent cell, hidden size 500, truncation $T = 200$, rollout horizon $k = 5$ unless stated, observation size 147 ($7 \times 7 \times 3$), action size 5. Training uses RMSprop ($\alpha = 0.95$, $\epsilon = 10^{-6}$) with four parameter groups at per-group learning rates $\lambda\sqrt{1/H}$ for $W_{\text{rec}}, W_{\text{out}}$, $\lambda\sqrt{1/(n_{\text{obs}} + n_{\text{act}})}$ for W_{in} and 0.1λ for the bias, with bias-only weight decay; $\lambda = 2 \times 10^{-3}$, batch size 8, gradient clipped per model to norm 1, 80,000 steps.

Arenas.

18×18 MiniGrid arenas (324 positions) with landmark layouts of C_1, C_2 and C_4 rotational symmetry. The observational discriminability index (ODI; the Spearman correlation between heading-averaged observation distance and spatial distance across all pairs of positions) is *not* matched across the three arenas, and cannot be: symmetric landmarks make distinct positions observationally identical, so discriminability necessarily falls with symmetry order. Measured over all 324 positions, ODI = 0.336 (C_1), 0.288 (C_2), 0.262 (C_4).

This does not confound the results, for two reasons. The decisive contrast, *axis* vs. *parity*, is made *within* one arena, on the same trajectories, so the observation stream and its ODI are identical between the two arms; the only difference is a 4×4 matrix on the heading block. And across arenas, ODI does not predict the readout: with a full compass, phase decoding is 0.981, 0.978 and 0.984 for C_1, C_2, C_4 (Pearson $r = -0.37$ against ODI, $p = 0.76$, $n = 3$ arenas), with the least discriminable arena scoring highest. Trajectories: 10,000 per arena, 200 steps, random start position and heading. The policy is a fixed-probability random walk (forward 0.60, turn left 0.15, turn right 0.15, stop 0.10), identical at every position, heading, and arena; nothing about it depends on which copy of the fundamental domain the agent occupies, which is what makes it measurable with respect to the agent's belief (Theorem).

Head-direction encodings.

The heading block of the action vector is left-multiplied by a fixed 4×4 matrix: identity (*full*); $\frac{1}{2}(I+P^2)$ where P is the cyclic shift (*axis*); the block-averaging matrix pairing $\{E, S\}$ and $\{W, N\}$ (*parity*); and $\frac{1}{4}\mathbf{1}\mathbf{1}^\top$ (*const*). Column sums are preserved, so activation magnitude is unchanged. Entropies under a uniform heading prior are $2, 1, 1, 0$ bits.

Ensemble training.

All networks sharing a computation graph (arena, encoding and seed all leave it unchanged; the horizon k does not) are trained as a single job using `torch.func.stack_module_state` with `vmap(functional_call)`, which lowers every per-model matrix multiply to one batched multiply and makes the kernel-launch count independent of the number of models. Forward passes are bit-exact against separate runs; gradients are bit-exact with noise disabled and agree to 2.3×10^{-10} absolute with the production noise on (`bmm` reassociation). Per-model gradient clipping, the four-group RMSprop and the sum-of-per-model-means loss are all preserved; each is checked against the reference implementation by test.

Orbit-phase decoding.

Hidden states are collected over 20,000 raw timesteps (no per-position averaging); repeated visits to one position therefore contribute multiple, slightly different rows, and `GroupKFold` groups every row from an orbit's two positions together regardless of how many visits each contributed. Orbits are $\{x, R^2x\}$ under the $180°$ rotation (or the four-element C_4 orbits where stated); the arena side is even, so no rotation has an integer fixed point and every orbit has exactly $|G|$ cells (162 orbits at C_2, 81 at C_4). Phase labels are balanced, and cross-validation folds hold out whole orbits, so the decoder must generalise a phase direction rather than memorise orbits or exploit a repeated visit to the same position appearing on both sides of a split. We report the accuracy of a logistic decoder, together with the R^2 of ridge decoders for the raw position and for the position within the fundamental domain. The latter is not a positive control (recovering a folded coordinate from an unfolded code requires a nonlinear map), so the sanity check that a code still represents space is max($R^2_{\text{raw}}, R^2_{\text{domain}}$).

We checked directly whether this pipeline is itself biased, since a 500-dimensional feature decoded by 5-fold `GroupKFold` over only 162 orbit groups is a regime where an improperly validated decoder could read above chance on a truly invariant code. The standardiser is fit on the training fold only (never on the pooled data before the split), and no other preprocessing step (no PCA, no whitening, no cross-validated regularisation strength) touches held-out rows before scoring, so there is no pre-split leakage in the code. We also ran the decoder on a synthetic code matched to this exact regime, 500 dimensions, 162 C_2 orbits, 20,000 rows, constructed to be exactly invariant to phase by hand: it reads 0.498, indistinguishable from the nominal 0.500. The decoder is therefore unbiased in the regime this paper uses it in. The $k{=}0$ full-compass condition's 0.536 and every folded network's 0.52–0.56 (Table 2) are not, consequently, a decoder artefact to be read against as an empirical floor; they are real signal in the trained networks themselves, a small residual above the nominal chance the permutation test above confirms is not noise. We attribute it, as in the Discussion, to finite training never reaching the exactly invariant solution a population-level, infinitely-trained network would.

Within-episode phase (Experiment 0).

The across-episode analysis above shows absolute orbit phase is undecodable in steady state, but that alone cannot distinguish a code that never carries phase information from one that carries it locally, right after an episode begins, and loses it as the trajectory accumulates. We tested this directly with a per-episode-relative label, $z(t) = \text{phase}(t) \oplus \text{phase}(0)$, decoded with a logistic model under `GroupKFold`-by-episode and `class_weight='balanced'`, reporting balanced accuracy (chance $= 0.5$ regardless of z's base rate) as a function of dwell time since episode start. An earlier version of this analysis, reporting raw accuracy, was invalidated by a majority-class confound: under a random walk, $P(z = 1)$ rises from about 0.08 near episode start to about 0.46 by $100+$ steps purely from position continuity, independent of the hidden state, so a majority-class-only classifier scored about 0.93 at 0–10 steps using zero real information; comparing every reported number against this baseline is now standard in the pipeline. The corrected result, $n = 10$ seeds each of *learned*, *axis* and *const* in the C_2 arena, is nominally above chance (Wilcoxon signed-rank against 0.5, exact two-sided, uncorrected) at every dwell bin tested for all three encodings, decaying from 0.66–0.74 near episode start toward the across-episode baseline (0.52–0.53) by $100+$ steps. The effect at the longest bin is markedly weaker for *learned* ($p = 0.049$, at the edge of detection) than for *axis*/*const* ($p \leq 0.0098$), though none actually falls to non-significance. The fold is therefore decodable-but-drifting: the network keeps a locally coherent orbit-frame for a window after an episode starts, which attenuates toward the steady-state floor rather than vanishing at the first step (an instantaneous fold) or never appearing at all (a code with no phase information to lose).

Isotypic spectrum.

Each unit's rate map is decomposed into the four characters of C_4; P_k is the normalised power in character k, $\sum_k P_k = 1$, and $P_1 = P_3$ for real maps. RA $= P_0 - P_2$ and odd $= P_1 + P_3$.

Compartment mazes.

Room-interior observational identity (two and four rooms) is verified by direct enumeration: rendering every room-interior cell and heading and comparing pixel images across rooms gives zero mismatch (0/144 pairs, two rooms; 0/432, four rooms). The connecting corridor is not: because every room's corridor opens onto one shared junction, a room at either end of the junction's span sees it truncated on one side while an interior room does not, and once the agent's forward view (F cells) reaches far enough into the corridor to include the junction, this asymmetry becomes visible (maximum single-channel pixel mismatch of 5/255, first appearing exactly where the view depth reaches the junction column). Room-interior views therefore satisfy the aliasing this maze is built to test; the maze as a whole, corridor included, does not instantiate a finite group acting freely on the full state space the way the rotationally symmetric C_2/C_4 arenas do, so the two- and four-room results are an empirical demonstration of the same principle rather than a direct instance of the Theorem. Room decoding (`label_rooms`) is restricted to room-interior timesteps for exactly this reason.

The two-room arrangements compound this: the corridor connecting the compartments is not the same length in both. The translation arrangement's doors are joined by a direct link (27 cells, door to door); the rotation arrangement's link runs around three edges of the arena to avoid cutting through the rotated room's wall ring (57 cells; 58 non-room cells in total against 28 for translation). A longer, more distinctive corridor is a channel by which recurrent memory of the just-completed transit, not the compass, could carry room identity into the following room-interior states, which would confound the translation/rotation contrast with corridor length rather than isolating head-direction equivariance. We tested this directly on the trained networks by conditioning room decoding on the number of steps since the animal last crossed into its current room. The two arrangements show opposite temporal signatures. Under translation, decoding is above chance immediately on entry (0.63 within the first two steps) and decays to the near-chance floor within about ten steps (0.52, matching the steady-state value reported in the main text); a corridor-memory trace is real but small and transient. Under rotation, decoding is *below* its ceiling immediately on entry (0.84) and *rises* to it over the following several steps (0.92, 0.98, 0.999, 1.00 at 1–2, 2–5, 5–10 and $10+$ steps). A corridor-memory account predicts the opposite ordering, highest immediately after the longer transit and decaying, so it does not explain the rotation result; the rise instead matches a network that needs a few steps of position-and-heading integration after entry to resolve which room it is in, consistent with the compass account this maze is built to test. The corridor-length asymmetry is therefore a real, small, and transient confound specific to the translation arrangement, not an alternative explanation for the dissociation itself: it makes the reported translation number (averaged over all room-interior timesteps, including the contaminated first few) a slightly conservative estimate of the steady-state fold, and it does not touch the rotation result at all.

CA1 reanalysis.

From ***Hockeimer et al.*** (***2023***) (JHU repository doi:10.7281/T15HMQD4, `AlleySuperpopDirVisitFiltered.csv`) we took, for each place field, its per-visit firing rates split by travel direction, and formed a directional index DI $= (r_+ - r_-)/(r_+ + r_-)$ with $+/-$ the two directions of the field's alley (E/W for horizontal alleys, N/S for vertical); fields with fewer than three visits in either direction were dropped. The primary statistic is the Pearson correlation over within-cell same-orientation field-DI *pairs* (a cell's repeating fields lie in translation-related alleys). The shuffle null permutes DI values across all repeating fields *within an orientation*, breaking the assignment of DIs to cells while preserving the marginal DI distribution, and is recomputed 2000 times. This null, however, also destroys cell-level directionality, so we ran two further controls that hold it fixed. First, the same correlation on *non-repeating* cells' same-orientation fields (which the fold does not predict should correlate). Second, a directionality-matched scramble that preserves each cell's per-orientation mean DI and resamples the within-cell residuals from the global pool, isolating the between-cell (global-directionality) contribution; the directional index is substantially a between-cell property (mixed-effects ICC 0.31), and this scramble reproduces (indeed exceeds) the observed correlation. The observed correlation is therefore consistent with the quotient but not separable from a conjunctive place-by-direction code in these data; we report it as such.

Statistics.

All contrasts are two-sided exact Mann–Whitney U tests on per-network values. The planned contrasts are *axis* vs. *parity* within each arena, at $n = 10$ per cell for s1 and s2 and $n = 8$ for s4; at $n = 10$

vs. 10 the exact two-sided floor is $2/\binom{20}{10} = 1.1 \times 10^{-5}$, and a p-value at that floor indicates complete separation (every network in one group above every network in the other). Bonferroni correction across the three planned arena contrasts is applied where stated. Secondary experiments carry their own n: the prediction-horizon and noisy-compass sweeps use $n = 6$ per cell (floor 0.00216), the isotypic re-run $n = 5$ (floor 0.0079). Folded decoding accuracies are compared to their $1/|G|$ floor by a one-sample t-test where a residual is claimed. No results were excluded; models failing a criterion are reported as such rather than dropped.

Data and code availability

The source data underlying every figure and table are provided as comma-separated files in `project5_sym` a per-file index maps each reported value to its source (`Report/data/README.md`). All code, the analysis pipeline that regenerates those files from trained checkpoints, and the figure-generation scripts are openly available at https://github.com/RavulapalliManas/Thesis_work (`project5_symmetry/`), and will be archived on Software Heritage on acceptance. The test suite pins the claims that fail silently: the bit-exactness of the batched ensemble against per-model training, that the $k = 0$ target is exactly the anchor observation, that a synthetic C_2-folded code reads chance phase while a code with no spatial signal is distinguishable from it, and that the isotypic identity RA $= P_0 - P_2$ holds. Trained checkpoints and raw trajectories (∼40 GB) are available from the author on request; every published number is reproducible from the committed data files without them.

Author contributions

M.V.S.R. designed the study, implemented the models and analyses, ran the experiments and wrote the manuscript.

Competing interests

The author declares no competing interests.

Use of AI tools

Large language models were used for code review, refactoring, and drafting assistance. All experiments, statistics and scientific claims were specified, executed and verified by the author; every number reported here is produced by the archived analysis scripts.

References

Burak Y, Fiete IR. Accurate path integration in continuous attractor network models of grid cells. PLoS Computational Biology. 2009; 5(2):e1000291. doi: 10.1371/journal.pcbi.1000291.

Calton JL, Stackman RW, Goodridge JP, Archey WB, Dudchenko PA, Taube JS. Hippocampal Place Cell Instability after Lesions of the Head Direction Cell Network. The Journal of Neuroscience. 2003; 23(30):9719–9731. doi: 10.1523/JNEUROSCI.23-30-09719.2003.

Carpenter F, Manson D, Jeffery K, Burgess N, Barry C. Grid cells form a global representation of connected environments. Current Biology. 2015; 25(9):1176–1182.

Castro PS. Scalable methods for computing state similarity in deterministic Markov decision processes. In: *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 34; 2020. p. 10069–10076.

Derdikman D, Whitlock JR, Tsao A, Fyhn M, Hafting T, Moser MB, Moser EI. Fragmentation of grid cell maps in a multicompartment environment. Nature Neuroscience. 2009; 12(10):1325–1332.

Diba K, Buzsáki G. Forward and reverse hippocampal place-cell sequences during ripples. Nature Neuroscience. 2007; 10(10):1241–1242.

Ferns N, Panangaden P, Precup D. Metrics for finite Markov decision processes. In: *Proceedings of the 20th Conference on Uncertainty in Artificial Intelligence (UAI)*; 2004. p. 162–169.

Foster DJ, Wilson MA. Reverse replay of behavioural sequences in hippocampal place cells during the awake state. Nature. 2006; 440(7084):680–683.

Gardner RJ, Hermansen E, Pachitariu M, Burak Y, Baas NA, Dunn BA, Moser MB, Moser EI. Toroidal topology of population activity in grid cells. Nature. 2022; 602(7895):123–128. doi: 10.1038/s41586-021-04268-7.

George D, Rikhye RV, Gothoskar N, Guntupalli JS, Dedieu A, Lázaro-Gredilla M. Clone-structured graph representations enable flexible learning and vicarious evaluation of cognitive maps. Nature Communications. 2021; 12(1):2392.

Givan R, Dean T, Greig M. Equivalence notions and model minimization in Markov decision processes. Artificial Intelligence. 2003; 147(1-2):163–223.

Grieves RM, Jenkins BW, Harland BC, Wood ER, Dudchenko PA. Place Field Repetition and Spatial Learning in a Multicompartment Environment. Hippocampus. 2016; 26(1):118–134. doi: 10.1002/hipo.22496.

Hartley T, Burgess N, Lever C, Cacucci F, O'Keefe J. Modeling place fields in terms of the cortical inputs to the hippocampus. Hippocampus. 2000; 10(4):369–379. doi: 10.1002/1098-1063(2000)10:4<369::AID-HIPO3>3.0.CO;2-0.

Hockeimer W, Lai RY, Natrajan M, Snider W, Knierim JJ, Giocomo LM. Leveraging place field repetition to understand positional versus nonpositional inputs to hippocampal field CA1. eLife. 2023; 12:e85599. doi: 10.7554/eLife.85599.

Kjelstrup KB, Solstad T, Brun VH, Hafting T, Leutgeb S, Witter MP, Moser EI, Moser MB. Finite scale of spatial representation in the hippocampus. Science. 2008; 321(5885):140–143.

Knierim JJ, Kudrimoti HS, McNaughton BL. Place cells, head direction cells, and the learning of landmark stability. The Journal of Neuroscience. 1995; 15(3):1648–1659. doi: 10.1523/JNEUROSCI.15-03-01648.1995.

Langston RF, Ainge JA, Couey JJ, Canto CB, Bjaalie JG, Witter MP, Moser EI, Moser MB. Development of the Spatial Representation System in the Rat. Science. 2010; 328(5985):1576–1580. doi: 10.1126/science.1188210.

Leutgeb S, Leutgeb JK, Barnes CA, Moser EI, McNaughton BL, Moser MB. Independent codes for spatial context and spatial location in the hippocampal formation. Science. 2005; 309(5734):619–623. doi: 10.1126/science.1110711.

Levenstein D, Efremov A, Eyono RH, Peyrache A, Richards BA. Sequential predictive learning is a unifying theory for hippocampal representation and replay. bioRxiv. 2024; doi: 10.1101/2024.04.28.591528, posted 29 April 2024.

Li L, Walsh TJ, Littman ML. Towards a unified theory of state abstraction for MDPs. In: *Proceedings of the 9th International Symposium on Artificial Intelligence and Mathematics*; 2006. p. 531–539.

Markus EJ, Qin YL, Leonard B, Skaggs WE, McNaughton BL, Barnes CA. Interactions between location and task affect the spatial and directional firing of hippocampal neurons. Journal of Neuroscience. 1995; 15(11):7079–7094.

McNaughton BL, Barnes CA, O'Keefe J. The contributions of position, direction, and velocity to single unit activity in the hippocampus of freely-moving rats. Experimental Brain Research. 1983; 52(1):41–49.

Momennejad I, Russek EM, Cheong JH, Botvinick MM, Daw ND, Gershman SJ. The successor representation in human reinforcement learning. Nature Human Behaviour. 2017; 1(9):680–692. doi: 10.1038/s41562-017-0180-8.

van der Pol E, Worrall D, van Hoof H, Oliehoek F, Welling M. MDP homomorphic networks: Group symmetries in reinforcement learning. In: *Advances in Neural Information Processing Systems*, vol. 33; 2020. p. 4199–4210.

Ravindran B, Barto AG. An algebraic approach to abstraction in reinforcement learning. PhD thesis, University of Massachusetts Amherst. 2004; .

Rich PD, Liaw HP, Lee AK. Large environments reveal the statistical structure governing hippocampal representations. Science. 2014; 345(6198):814–817.

Sorscher B, Mel GC, Ocko SA, Giocomo LM, Ganguli S. A unified theory for the computational and mechanistic origins of grid cells. Neuron. 2023; 111(1):121–137.e13. doi: 10.1016/j.neuron.2022.10.003.

Spiers HJ, Hayman RMA, Jovalekic A, Marozzi E, Jeffery KJ. Place Field Repetition and Purely Local Remapping in a Multicompartment Environment. Cerebral Cortex. 2015; 25(1):10–25. doi: 10.1093/cercor/bht198.

Stachenfeld KL, Botvinick MM, Gershman SJ. The hippocampus as a predictive map. Nature Neuroscience. 2017; 20(11):1643–1653. doi: 10.1038/nn.4650.

Taube JS, Muller RU, Ranck JB. Head-direction cells recorded from the postsubiculum in freely moving rats. I. Description and quantitative analysis. The Journal of Neuroscience. 1990; 10(2):420–435. doi: 10.1523/JNEUROSCI.10-02-00420.1990.

Whitehead SD, Ballard DH. Learning to perceive and act by trial and error. Machine Learning. 1991; 7(1):45–83.

Whittington JCR, Muller TH, Mark S, Chen G, Barry C, Burgess N, Behrens TEJ. The Tolman-Eichenbaum Machine: Unifying space and relational memory through generalization in the hippocampal formation. Cell. 2020; 183(5):1249–1263.e23. doi: 10.1016/j.cell.2020.10.024.

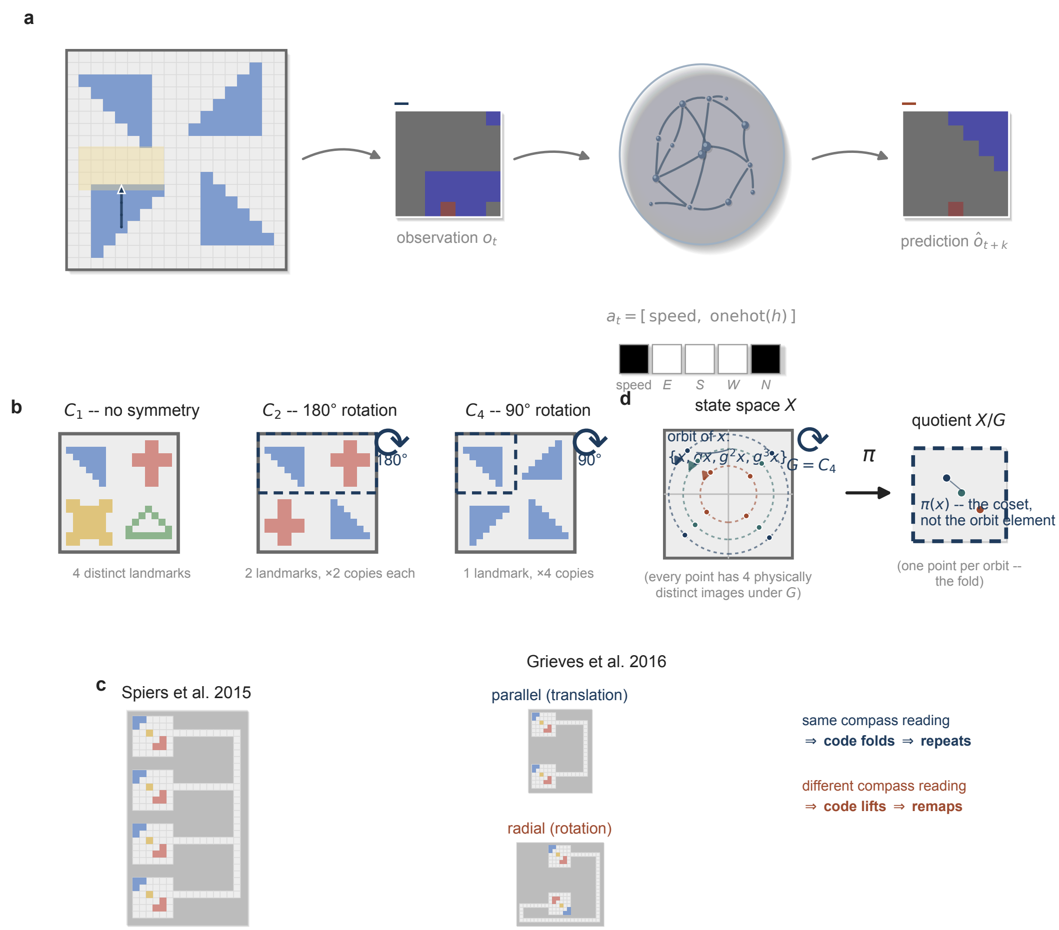

Figure 1. Overview. (**a**) The pipeline: an agent moves through a symmetric arena under actions drawn from a five-dimensional SpeedHD vector (Methods); a predictive RNN (pRNN) receives the current $7 \times 7 \times 3$ egocentric observation o_t and predicts the observation k steps ahead, $\hat{o}_{t+k}$. (**b**) The three arenas used throughout, named by their landmark symmetry group: C_1 (four distinct landmarks, no symmetry), C_2 (one landmark pair repeated under 180° rotation), and C_4 (one landmark repeated four times under 90° rotation); dashed outline marks a fundamental domain. (**c**) The animal phenomenon this study explains: ***Spiers et al.*** (***2015***) recorded a high degree of spatial repetition across four visually identical compartments in a row; ***Grieves et al.*** (***2016***) showed the same compartments arranged in parallel produce repeated place fields, while arranging them radially, so that a rotation rather than a translation relates them, produces remapping. (**d**) The central object of the Theorem: a group G (here C_4) partitions the state space X into orbits; the quotient map π sends each orbit to a single point of X/G, discarding which physically distinct member of the orbit the agent occupies.

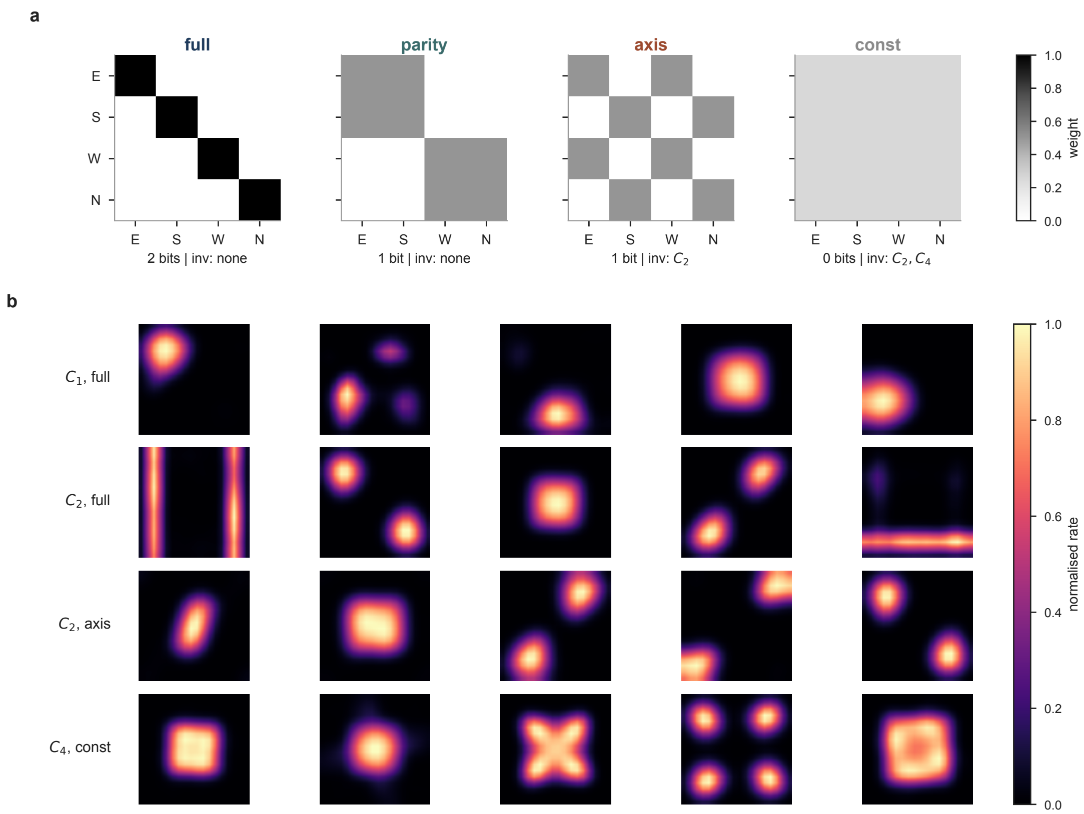

Figure 2. Design and phenotype. (**a**) The four head-direction encodings, as 4×4 transforms of the one-hot heading block {E,S,W,N}; grey level is the matrix weight. *full* is the identity (two bits); *parity* and *axis* collapse the compass to one bit with different partitions; *const* discards it. *axis* and *const* are invariant under the $180°$ rotation and fold the C_2 arena; *const* is the only encoding invariant under the full C_4 group. *axis* and *parity* carry identical entropy (one bit) and differ only in invariance. (**b**) Mean firing-rate maps (smoothed, peak-normalised) of example units. Fields are single and sharp when the code does not fold, and repeat at the symmetry-related locations, once under C_2 (*axis* encoding) and four times under C_4 (*const*). Units are representative; the complete unselected set for each condition is given in the Supplementary Information.

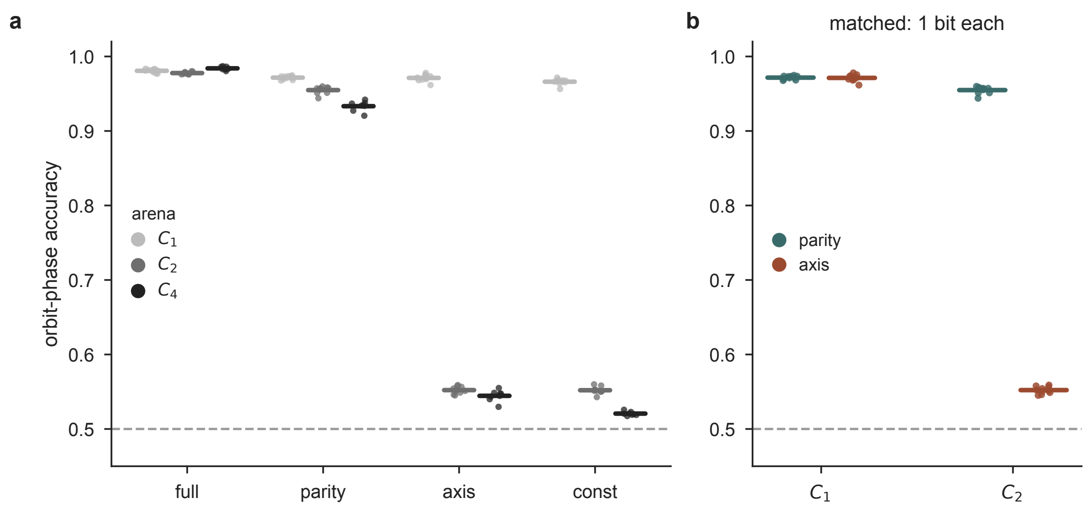

Figure 3. Invariance, not information, folds the code. (**a**) Orbit-phase decoding accuracy for every trained network (one dot per network; horizontal bar, group mean), by head-direction encoding and arena; chance 0.5 (dashed). *full* and *parity* stay high in every arena, whereas the invariant encodings *axis* and *const* fall to chance in the symmetric arenas. (**b**) The matched-entropy contrast: *axis* and *parity* carry one bit each and are indistinguishable in the C_1 arena (both ≈ 0.97) but separate completely in C_2 (*parity* 0.955, *axis* 0.552; every *parity* network scores above every *axis* network, Mann–Whitney $p = 1.1 \times 10^{-5}$). Data of Table 2.

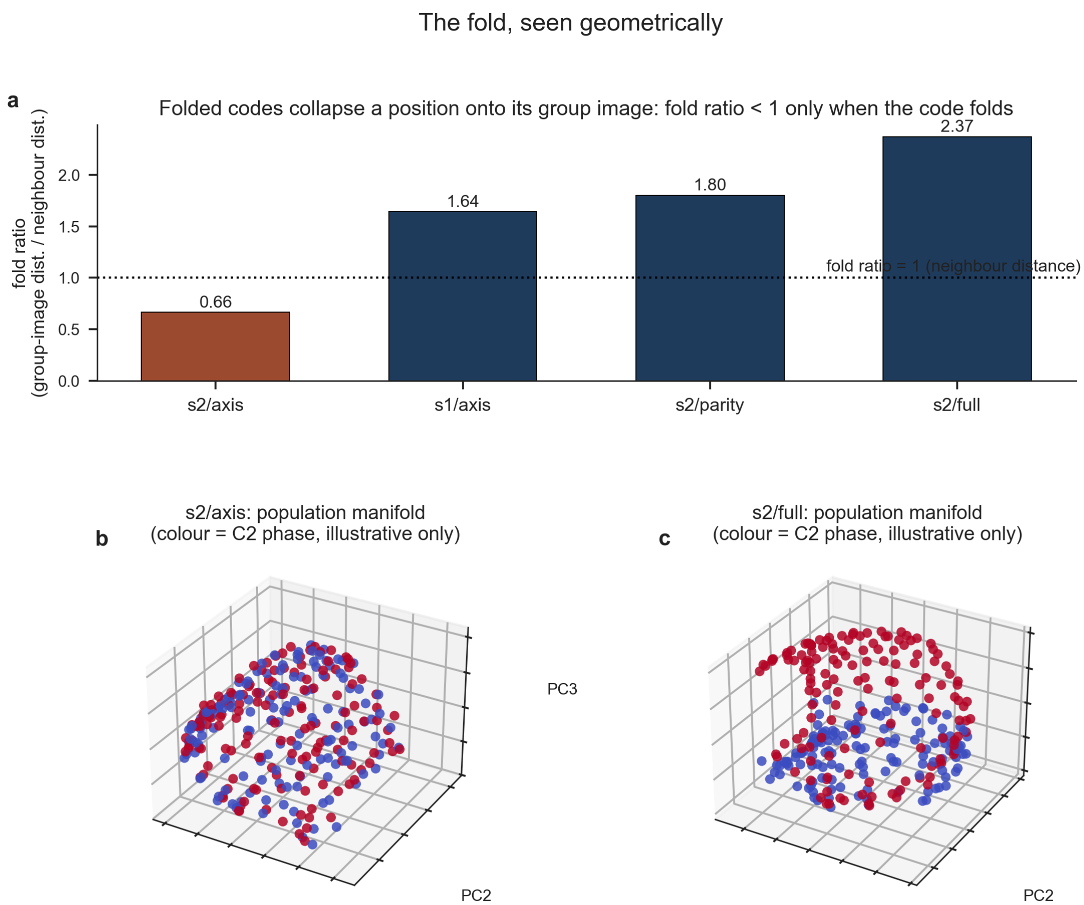

Figure 4. The fold, seen geometrically. (**a**) The fold ratio, distance between a cell and its $180°$ image in units of the distance to a spatial neighbour, for one representative network per condition: below one only for *axis* in the C_2 arena (0.66), the sole folded case shown, and above one for *axis* in the C_1 arena (1.64, same encoding, no symmetry to fold onto), *parity*/C_2 (1.80) and *full*/C_2 (2.37). The ratio is a property of the full hidden space, not of the linear projection used to plot it below: the folded case reads below one and every non-folded case above one under distance-preserving nonlinear embeddings (Isomap, t-SNE) as well as PCA. (**b**, **c**) Position-conditioned population manifold of the most- and least-folded networks shown in (a) (one point per arena cell, PCA to three dimensions, coloured by C_2 orbit phase), illustrative only: the fold ratio in (a) is the quantitative claim. Phases superimpose under *axis*/C_2 and occupy separate regions under *full*/C_2. We reduce with PCA because the claim is metric, two positions coincide — and unlike neighbour-graph embeddings PCA preserves distances rather than reshaping them.

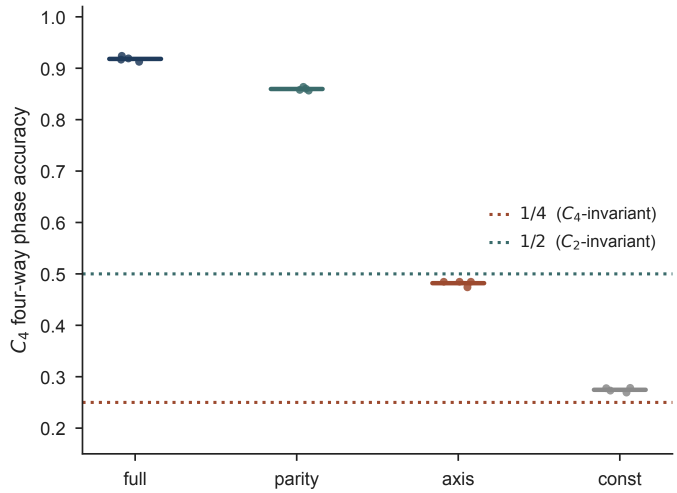

Figure 5. The C_4 arena reproduces the group-theoretic ceiling. Four-way orbit-phase accuracy in the C_4 arena (one dot per network; bar, mean; chance 0.25). A code folded onto X/G can name the coset but not the orbit element, so a $|G|$-way decoder is bounded by $1/|G|$. *const*, invariant under the full C_4 group, falls to the $1/4$ ceiling (dotted); the C_2-invariant *axis* falls to the $1/2$ ceiling; *full* and *parity*, not invariant, stay high. Ceilings are the group-theoretic predictions, not fits.

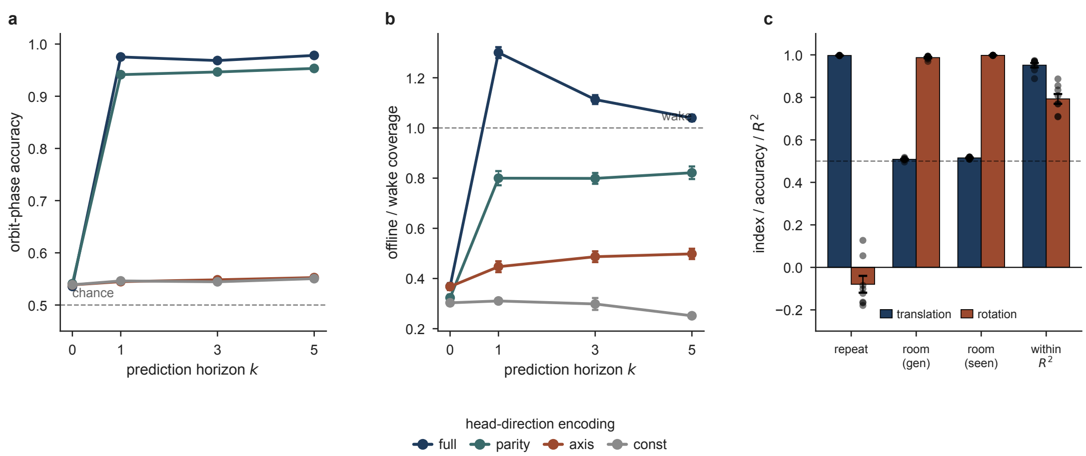

Figure 6. Functional consequences of the fold. (**a**) Orbit-phase decoding accuracy vs prediction horizon k in the C_2 arena (chance 0.5); the non-invariant encodings (*full*, *parity*) resolve the fold at $k \geq 1$, the invariant ones (*axis*, *const*) stay near chance. (**b**) Coverage of the self-generated offline trajectory as a fraction of a wake path of equal length (dashed line, wake). Both step up at $k = 1$ and are ordered by how much of the compass the encoding keeps; points are mean ± s.e.m., $n = 6$. (**c**) Place-field repetition across two observationally identical compartments ($n = 8$ each): under translation the fields repeat and the room is undecodable (chance 0.5), under rotation the fields do not repeat and the room decodes; within-room position (R^2) stays high in both. Dots are individual networks.

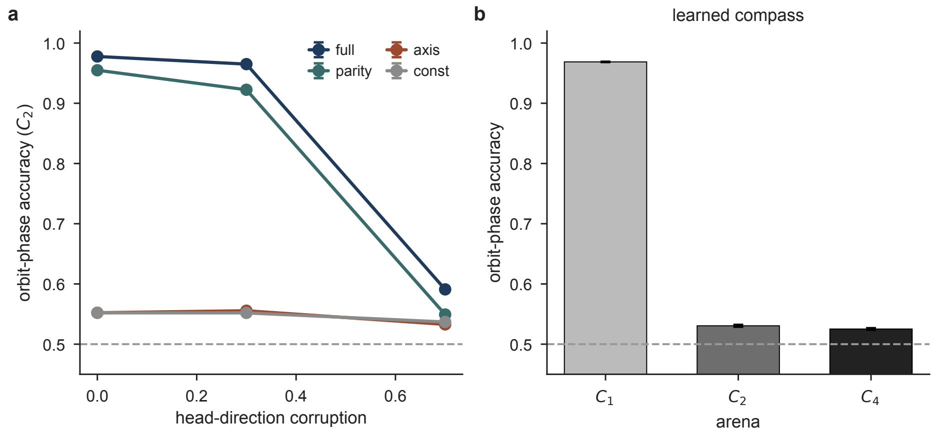

Figure 7. The quotient survives a degraded and a learned compass. (**a**) HD-lesion dose-response: orbit-phase accuracy in the C_2 arena as a function of the fraction of heading steps randomised (chance 0.5, dashed). The non-invariant encodings (*full*, *parity*) degrade smoothly, while the invariant ones (*axis*, *const*) stay pinned at chance. (**b**) A learned, angular-velocity compass, by arena: it resolves C_1 (0.97) but folds C_2 and C_4 to chance, building the quotient with no imposed encoding.

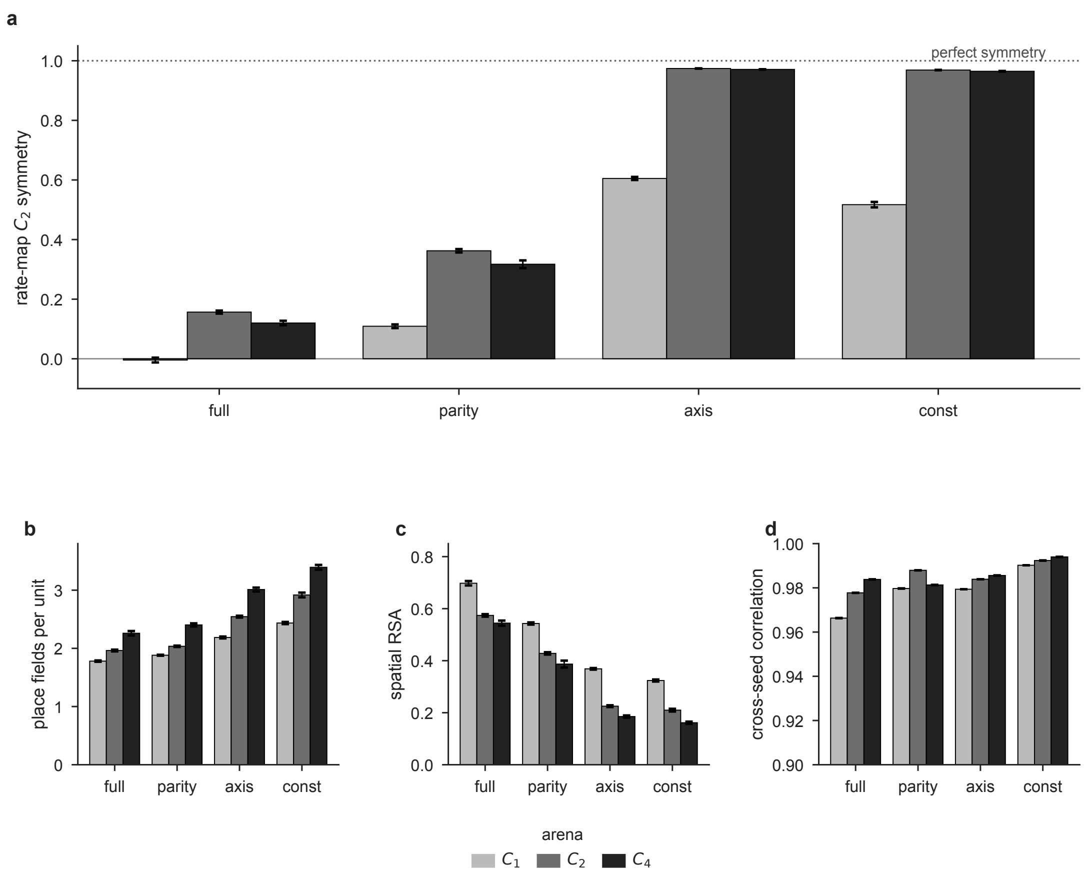

Figure 8. Quantifying the fold. (**a**) Rate-map C_2 symmetry index (correlation of a unit's map with its $180°$ rotation), the fold-specific readout; it saturates near 0.97 only when the encoding is invariant to the arena's symmetry, up to the perfect-symmetry line at 1. (**b**) Mean place fields per unit; rises with both arena symmetry and encoding invariance, but also in the C_1 arena where nothing folds, so field count alone is confounded. (**c**) Spatial representational similarity falls as the encoding is ablated and as arena symmetry increases, the signature of collapse onto the quotient. (**d**) Cross-seed correlation between independently trained networks stays above 0.96 throughout (y zoomed to 0.9–1.0): the fold is reproducible, not degenerate. Arenas $C_1/C_2/C_4$; bars are mean ± s.e.m. over networks ($n = 10$ per encoding for C_1 and C_2, $n = 8$ for C_4).

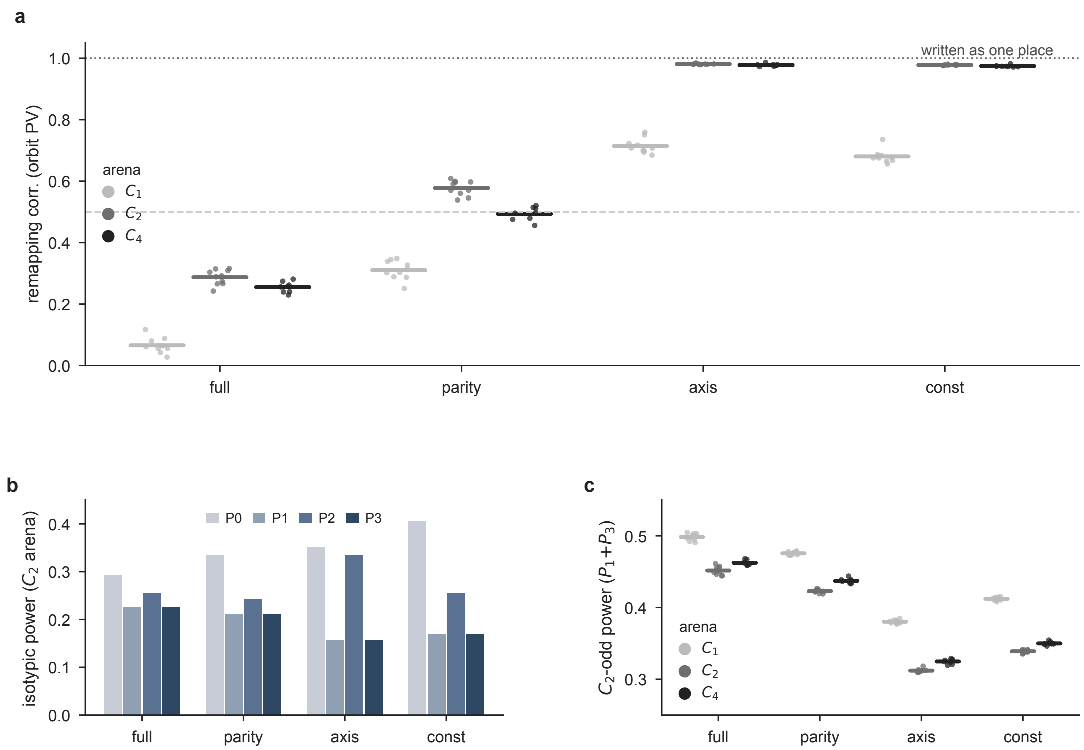

Figure 9. The quotient in the population code. (**a**) Remapping, read as the population-vector correlation between a position and its symmetry image (***Leutgeb et al., 2005***) (one dot per network; bar, mean). Under the invariant encodings (*axis*, *const*) the correlation reaches ≈ 0.98 in the C_2 and C_4 arenas, the two copies are written as one place, while *full* separates them (≈ 0.28) and the matched *parity* is intermediate (0.58 in C_2). (**b**) Isotypic power spectrum (C_2 arena): the invariant encodings concentrate power in the even characters (P_0, P_2) and suppress the odd ones (P_1, P_3), the representation-theoretic content the encoding's invariance dictates. (**c**) C_2-odd power P_1+P_3 by encoding and arena: lower for the invariant encodings and falling into the symmetric arenas, the descriptive correlate of the fold. Arenas $C_1/C_2/C_4$.

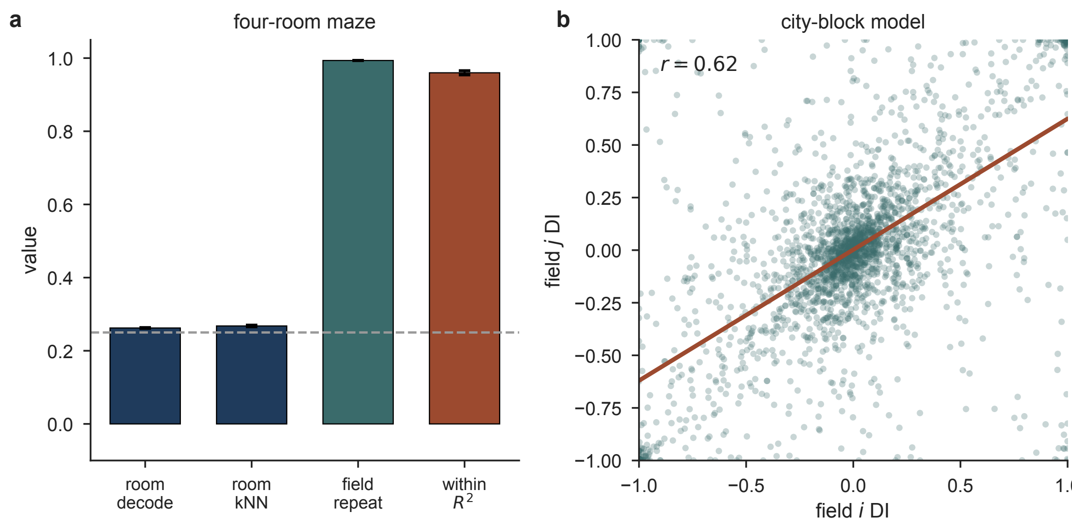

Figure 10. The predictive model reproduces place-field repetition. (**a**) The four-room maze of ***Spiers et al.*** (***2015***), model ($n = 8$): four translation-related identical rooms fold onto one map. The room decodes at chance (0.25, dashed), by a generalising decoder and by a nearest-neighbour classifier at cells it has already seen, while fields repeat across the rooms (repetition 0.99) and within-room position stays intact ($R^2 = 0.96$). (**b**) A predictive network trained on a city-block maze shows the directional-repetition signature, predicted before any comparison to animal data: the directional index of a field (i) against that of its same-orientation partner (j) is positively correlated ($r = 0.62$; scatter subsampled for display, statistic from all pairs). Line, least-squares fit. An exploratory, non-diagnostic reanalysis of real CA1 data testing the same prediction is in the Supplementary Information (Fig. S9).

Table 6. The model against the animal literature. "Reproduced" means the model shows the phenomenon without being fitted to it; "absent" means the phenomenon cannot exist in this architecture and we do not claim it; "tested, not found" means the architecture could in principle show the phenomenon but a direct test did not find it.

observation in animals	source	in this model	verdict
Fields repeat across translationally related identical compartments; path integration does not separate them	***Spiers et al.*** (***2015***)	translation leaves the compass invariant, so the code folds onto X/G	reproduced
Repetition in parallel compartments, markedly less in radial ones; survives removal of the orientation landmark; attributed to self-motion	***Grieves et al.*** (***2016***)	rotation makes the compass equivariant, so the code lifts	reproduced
Place fields repeat across the arms of a hairpin maze, and the repetition is running-direction dependent	***Derdikman et al.*** (***2009***)	folding onto X/G makes one unit fire at all $\|G\|$ orbit-mates, and the direction dependence is the compass entering the fold, but we have not built and run a hairpin-maze arena to test this directly	consistent
Head-direction input is required for a stable map; ADN lesions destabilise place fields	***Calton et al.*** (***2003***)	the *const* encoding folds the C_2 arena to near chance	consistent
Head direction plus recurrence plus prediction are needed for offline replay	***Levenstein et al.*** (***2024***)	offline trajectories exceed the coverage of a wake path of equal length only with the full compass and $k \geq 1$; coverage falls monotonically as the compass is ablated ($1.30 \rightarrow 0.80 \rightarrow 0.45 \rightarrow 0.31\times$ wake), and the autoencoder never leaves $0.37\times$. Replay appears at the same horizon as symmetry resolution and improves no further with a longer one	reproduced
A predictive objective is sometimes read as implying prospective (anticipatory) place coding	—	rebuilding each rate map on the position up to five steps ahead does not sharpen it at any prediction horizon; the network's predictive character shows up in offline replay, not in a spatial lead of its fields	tested, not found
Rate remapping: a slight rate-based discrimination between otherwise repeated fields	***Spiers et al.*** (***2015***)	the pure predictive model folds completely (repetition $+0.997$, compartment undecodable at 0.515, chance 0.5); a graded-cue sweep (Discussion) found no rate-based discrimination at any cue strength and no departure of repetition from ceiling; a small, chance-exceeding room-decode residual is present already at zero added cue	tested, not found
Grid cells on a toroidal population manifold, invariant across environments and sleep	***Gardner et al.*** (***2022***)	no grid cells (gridness ≤ 0.08); the torus cannot exist here	absent
Head-direction cells lie on a ring manifold that matures before place and grid coding	***Taube et al.*** (***1990***); ***Langston et al.*** (***2010***)	head direction is supplied as a four-point input, not learned	absent

Table 7. Where models of the spatial code stand on the symmetry dissociation. The decisive column is the last: the parallel and radial arrangements are locally identical, so any model that reads firing from local geometry or the current observation predicts the same repetition in both, and cannot produce the dissociation ***Grieves et al.*** (***2016***) observed.

model class	fields learned from experience	repetition in identical rooms	parallel-vs-radial dissociation
Continuous attractor / path integration (***Burak and Fiete, 2009***; ***Sorscher et al., 2023***)	metric built in	not addressed	no
Boundary-vector / geometry (***Hartley et al., 2000***)	partly	yes	no (predicts repetition in both)
Successor representation (***Stachenfeld et al., 2017***)	yes (predictive)	depends on the given state space	not addressed
Clone-structured graph (***George et al., 2021***)	yes (transition clones)	no (context splits the clones)	no (predicts *separation*)
Tolman–Eichenbaum machine (***Whittington et al., 2020***)	yes (structural)	yes, if the action code is symmetry-invariant	consistent
Reconstruction / autoencoder (this work, control)	yes	yes	no, folds both ($p = 0.59$)
Predictive RNN (***Levenstein et al., 2024***), this work	yes (emergent)	yes	**yes**, set by HD invariance